

Perturbation Response Audit

GAIRA Raman Reference Atlas v0.1 — linking perturbations to latent motifs

Atlas	NMF k=24, FROZEN (09ed804a4083..., verified unchanged)
Response fingerprints	53 serum-spike analytes, 24-D Δ-activation each
Dose experiments	6 ILS adenine arms + ergothioneine (component-resolved)
Loop closure	adenine → c3 (its own encoding component) in 4/6 arms
Purine sub-structure	recovered: {adenine,hypoxanthine} vs {xanthine,guanine}
Uricase depletion	selective loss in a purine-encoding component
Key caveat	theme LABELS unreliable on low-purity components; identities are not
Atlas modified?	No

PRINCIPAL FINDING

The loop closes at the level of component IDENTITY, not theme label. When adenine is perturbed, the single strongest-responding latent component is the one that actually encodes adenine in the reference atlas (c3) — reproducibly across substrates and lasers — even though the audit's coarse label for c3 is 'sterol'. Perturbation response is a sharper probe of component identity than the static audit labels, and it recovers a purine sub-classification the atlas was never given.

Executive summary

53 fingerprints · adenine→c3 in 4/6 arms · atlas frozen

Question

The Component Audit said what each latent Raman motif is; the Spike Validation said perturbations move through the atlas. This study connects them: when a known analyte is perturbed, which latent motifs respond, and does the response match the motif's chemistry? The atlas was frozen and verified unchanged.

Observed results

1. The response is selective, not a global amplitude shift. In the strongest adenine arm only 7 of 24 components rise while 17 fall; the two largest movers are c3 and c13, and c3's top reference analyte is adenine itself.
2. Loop closure at component identity. For adenine the single strongest-responding component is its own encoding component c3 in 4 of 6 substrate×laser arms — a reproducible closure across acquisition conditions.
3. Theme labels are the weak link, not the components. Theme-label agreement is only 15/128 responsive-component instances, because low-purity components carry misleading coarse labels (c3='sterol' but encodes adenine; c13='pyrimidine' but is thymine-dominated). The components are chemically real; the labels are not reliable.
4. Purine sub-structure recovered. Response fingerprints split the five purines into {adenine, hypoxanthine} (cos 0.88) and {xanthine, guanine} (cos 0.90), anti-correlated across the divide — the 6-oxo/amino versus 2,6-dioxo distinction, which the atlas was never told about.
5. Uricase depletion is selective. Enzymatic urate removal produces a targeted change in a purine-encoding component (c15 Δ -0.106) rather than a global shift — mechanistically consistent with removing a purine.
6. Response fingerprints recover chemical family better than raw spectra (ARI 0.146 vs 0.078), though both remain modest.

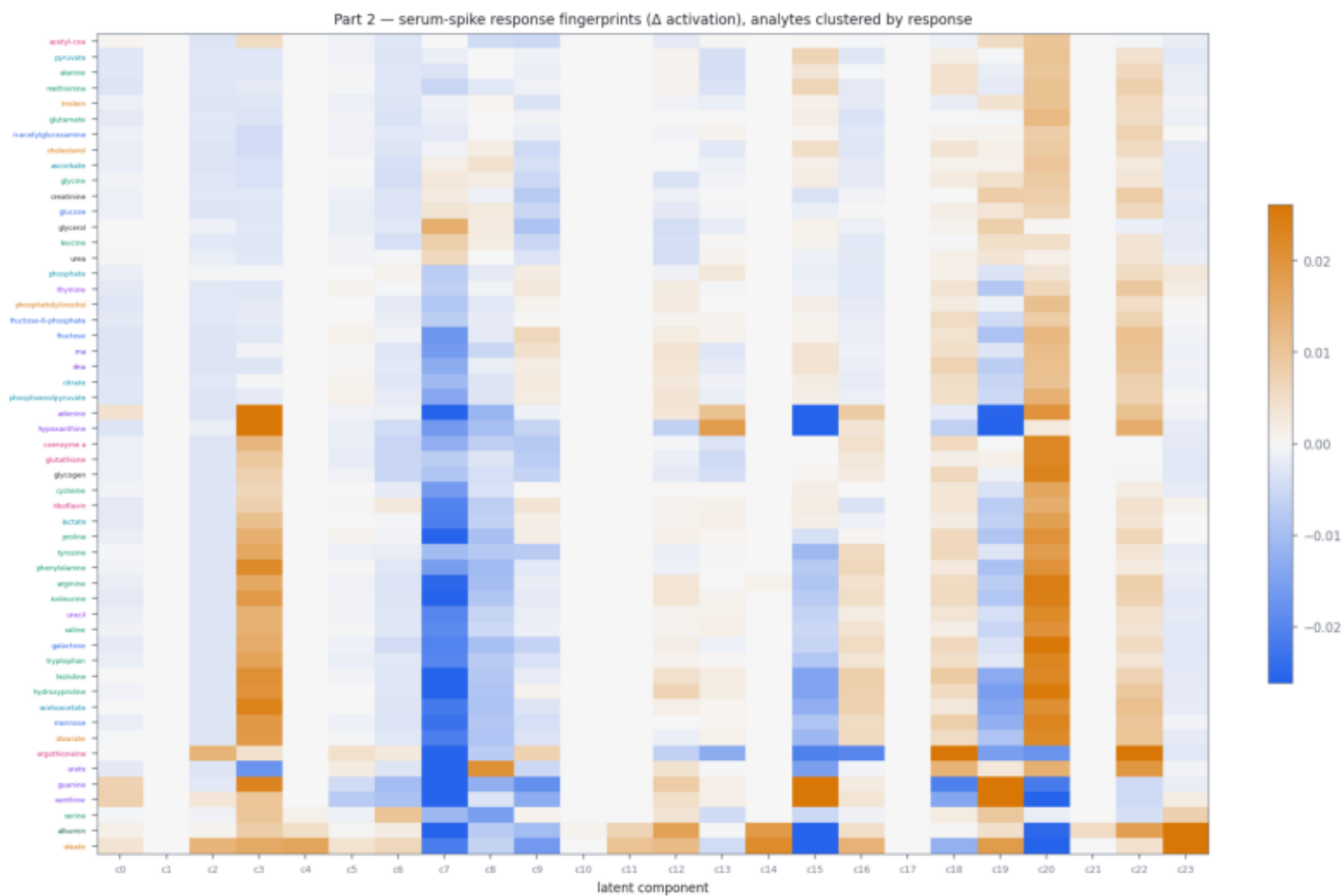
What the data do NOT support

7. Matrix-invariant signatures. Pure-analyte and serum-spike fingerprints for the same analyte agree only at median cosine $-0.012 \approx 0$: a component response is matrix-specific and must not be reused across matrices.
8. Identity recovery for weak adsorbers. Serum responders move more but not more focused (entropy unchanged); most non-purine spikes produce weak, non-specific activation, consistent with the prior finding that these Ag-SERS spectra are background-dominated.

Interpretation (separated from observation)

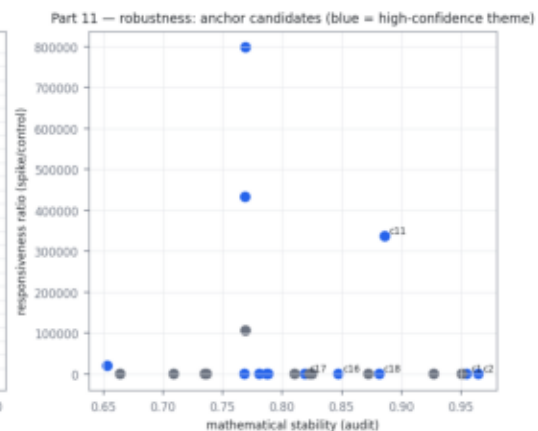
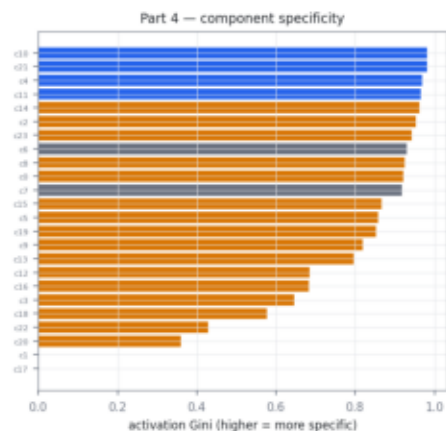
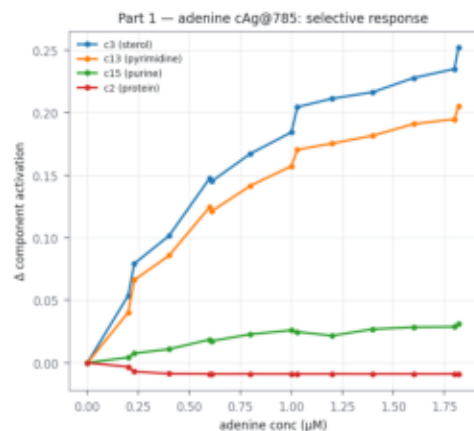
The atlas behaves as a coordinate system whose AXES ARE CHEMICALLY REAL but whose NAMES were assigned too coarsely by a static purity metric. Perturbation is the sharper naming tool: driving an analyte reveals which axis carries it. The immediate consequence for the ontology is to re-anchor low-purity component labels on their reference loadings and on perturbation identity, not on the dominant chemical family.

Part 2



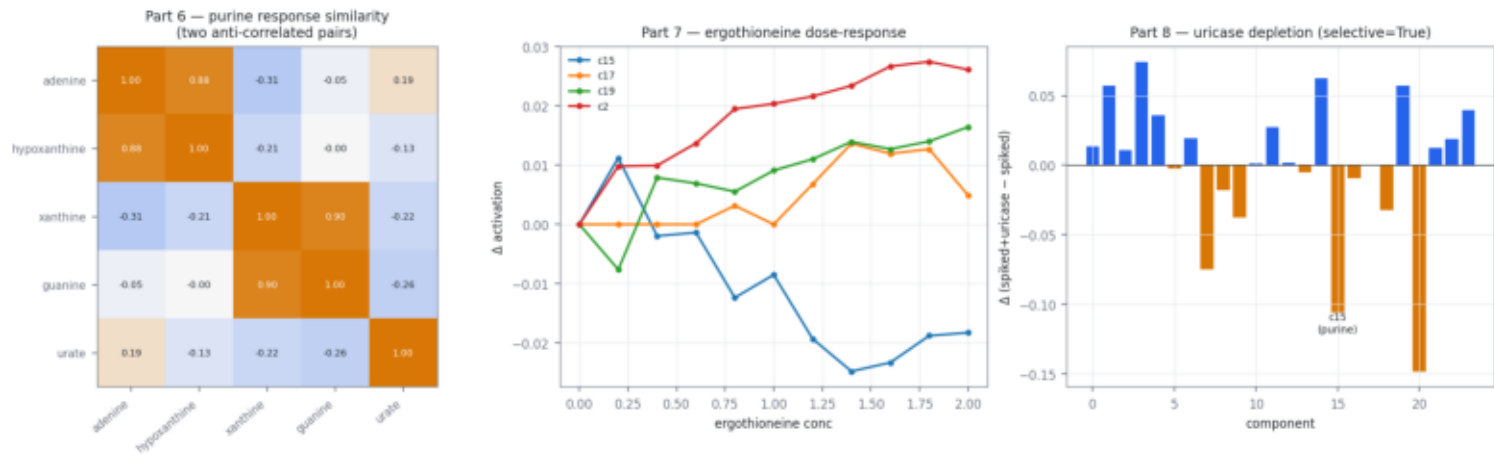
Part 2 — response fingerprints. Each row is one serum-spiked analyte's 24-component Δ -activation; analytes are clustered by their response and coloured by chemical family.

Part 1/4/11



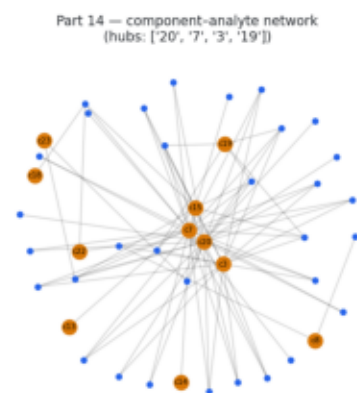
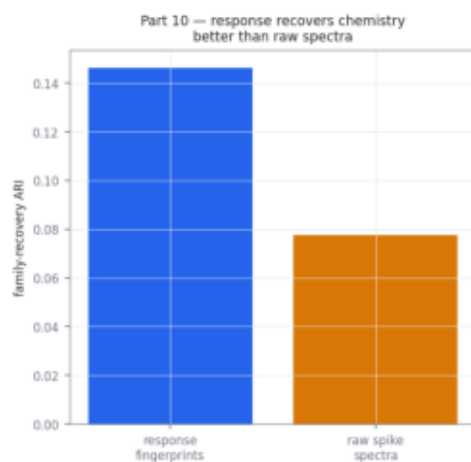
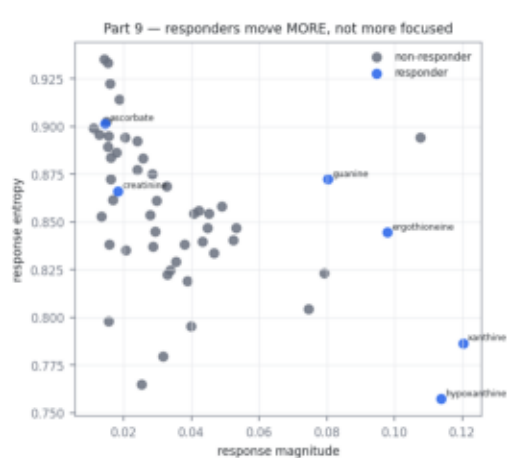
Part 1/4/11 — component dose-response for adenine (selective), per-component specificity, and the robustness map identifying BSV-anchor candidates.

Part 6/7/8



Part 6/7/8 — purine response similarity (two anti-correlated pairs), ergothioneine dose-response, and the selective uricase depletion.

Part 9/10/14



Part 9/10/14 — responders vs non-responders, family recovery vs raw spectra, and the component-analyte bipartite network with its hubs.

Methodology

Vocabulary discipline

A COMPONENT is a mathematical latent Raman motif (an NMF basis vector). Its THEME is a tentative post-hoc label from the Component Audit. A RESPONSE is the measured change in component activation under perturbation. A response that matches a theme corroborates it; a mismatch is reported, never hidden.

Projections

All 24-component coordinates are the frozen-atlas projections computed in the Spike Validation study (NMF dictionary held fixed). No spectrum was reprocessed and the atlas fingerprint was verified byte-identical before and after.

Identity test (the key methodological choice)

Because theme labels proved unreliable, the primary link test asks whether the perturbed analyte is itself among the top reference analytes that LOAD the responding component (from the audit's composition table) — a label-independent identity test. Synonyms and stereo-descriptors are canonicalised before matching.

Nulls and statistics

Component dose-response uses Spearman ρ with a saturating (Langmuir) vs linear model comparison; response fingerprints carry a 500-sample replicate bootstrap CI per component; family recovery is ARI against chemical family, compared head-to-head against clustering the raw spike spectra.

Datasets (all Ag/Au-SERS, out of domain for a Raman atlas)

ILS adenine 3381 · pure Ag-SERS 265 · serum spikes 265 (53 analytes) · serum baseline 15 · uricase 20 · isotopic 73 · ergothioneine 55

Response atlas

one page per serum-spiked analyte (53) — Part 13

xanthine

family: purine · RESPONDER · spike encodes: c15



Interpretation. The spike drives components c15, c20, c19 most strongly (themes: purine, saccharide, protein). Its own encoding component(s) c15 ARE among the strongest responders — an identity match. Pure-vs-serum consistency for this analyte is cos +0.74.

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

hypoxanthine

family: purine · RESPONDER · spike encodes: no atlas component



Interpretation. The spike drives components c3, c19, c15 most strongly (themes: sterol, protein, purine). This analyte does not correspond to any single atlas component, so no identity match is possible. Pure-vs-serum consistency for this analyte is $\cos +0.89$.

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

oleate

family: lipid · weak / non-responder · spike encodes: no atlas component

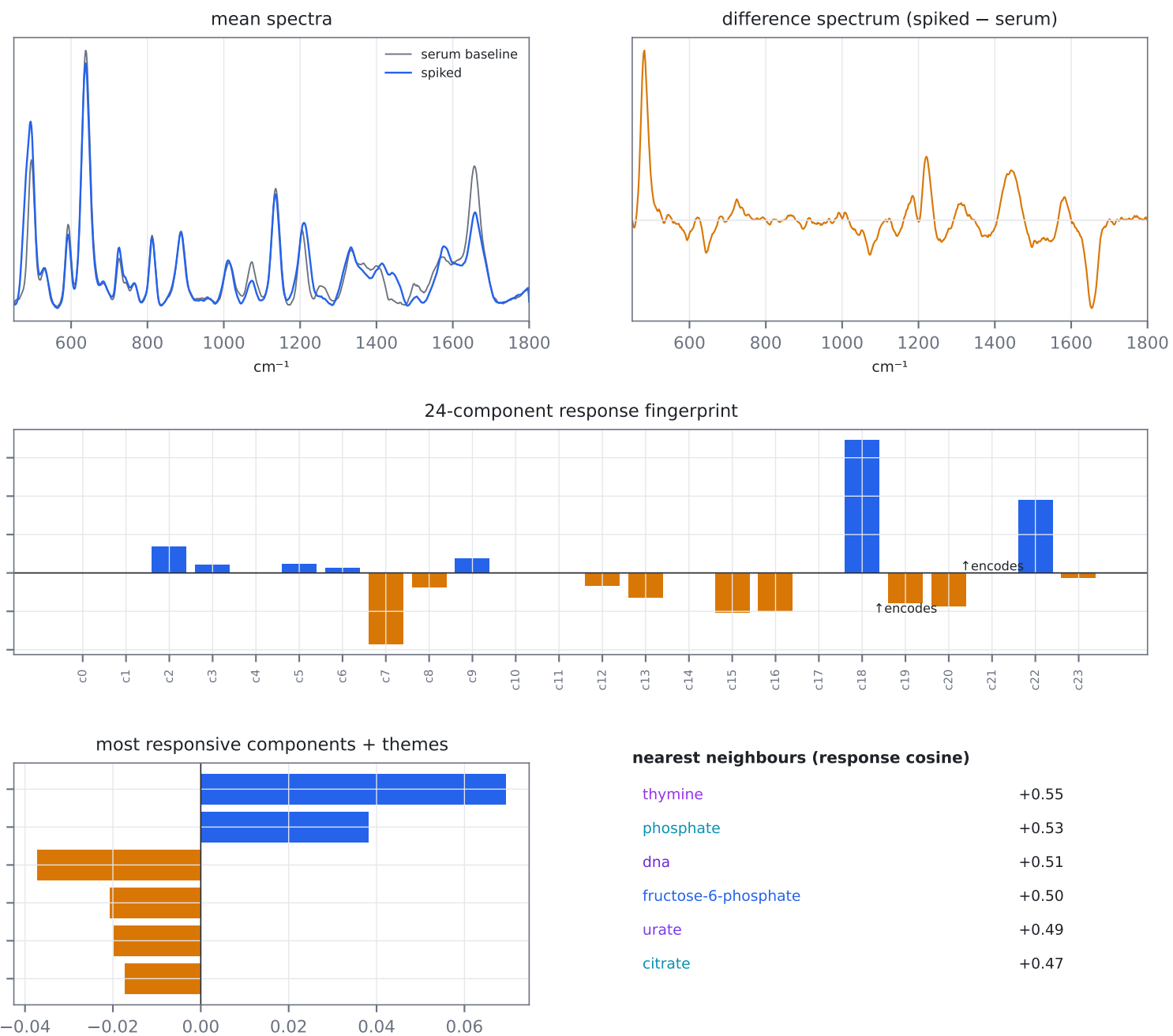


Interpretation. The spike drives components c15, c20, c23 most strongly (themes: purine, saccharide, organic_acid). This analyte does not correspond to any single atlas component, so no identity match is possible. Pure-vs-serum consistency for this analyte is cos -0.12 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

ergothioneine

family: cofactor · RESPONDER · spike encodes: c19, c21

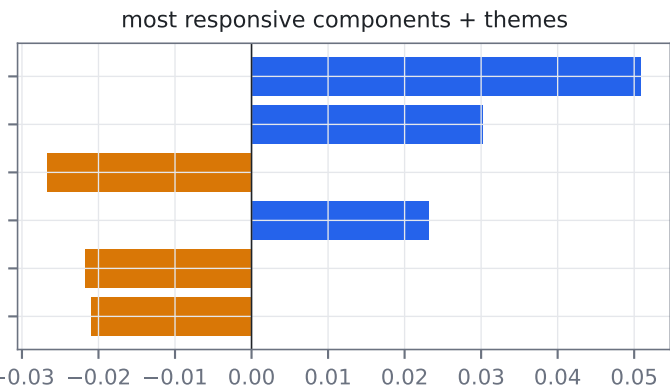
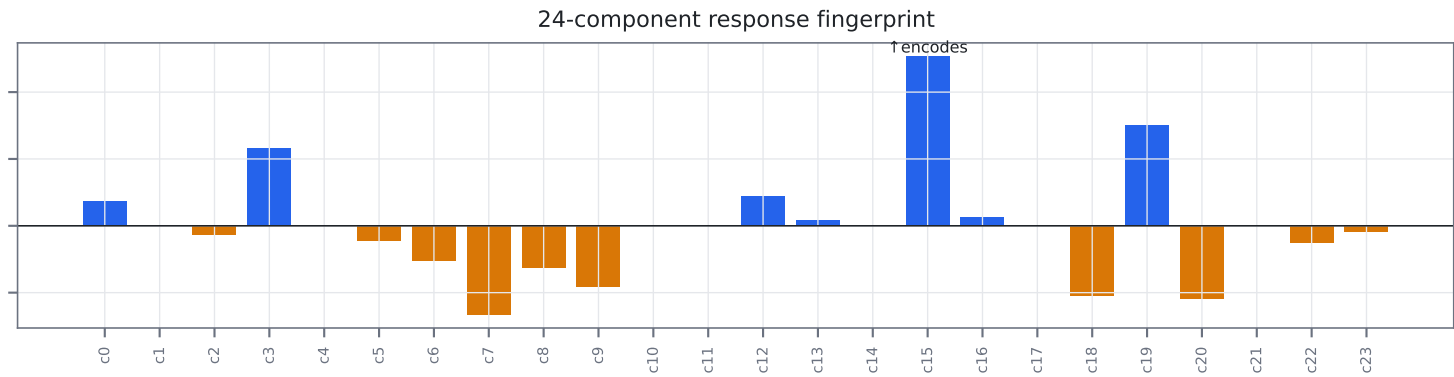
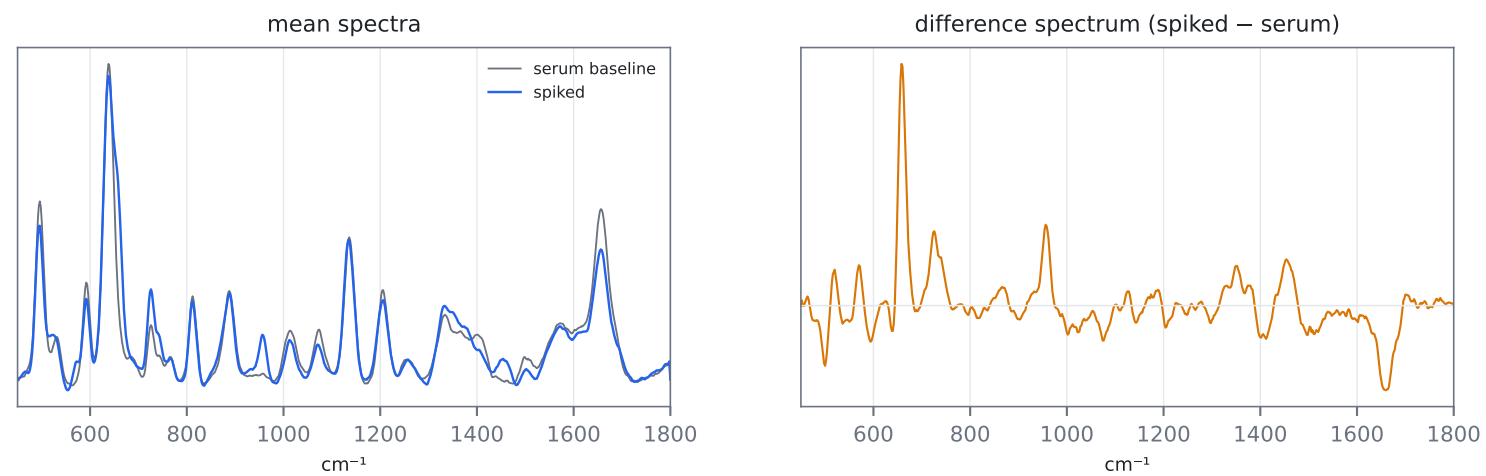


Interpretation. The spike drives components c18, c22, c7 most strongly (themes: saccharide, protein, protein). Its own encoding component(s) c19, c21 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is cos +0.53.

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

guanine

family: purine · RESPONDER · spike encodes: c15



nearest neighbours (response cosine)

xanthine	+0.90
serine	+0.28
acetyl-coa	+0.24
lactate	+0.13
coenzyme a	+0.12
cysteine	+0.10

Interpretation. The spike drives components c15, c19, c7 most strongly (themes: purine, protein, protein). Its own encoding component(s) c15 ARE among the strongest responders — an identity match. Pure-vs-serum consistency for this analyte is cos +0.40.

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

albumin

family: protein · weak / non-responder · spike encodes: no atlas component



Interpretation. The spike drives components c15, c7, c23 most strongly (themes: purine, protein, organic_acid). This analyte does not correspond to any single atlas component, so no identity match is possible. Pure-vs-serum consistency for this analyte is cos +0.05 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

adenine

family: purine · weak / non-responder · spike encodes: c3



Interpretation. The spike drives components c3, c7, c15 most strongly (themes: sterol, protein, purine). Its own encoding component(s) c3 ARE among the strongest responders — an identity match. Pure-vs-serum consistency for this analyte is cos +0.08 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

hydroxyproline

family: amino_acid · weak / non-responder · spike encodes: c11



Interpretation. The spike drives components c7, c20, c3 most strongly (themes: protein, saccharide, sterol). Its own encoding component(s) c11 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is cos -0.15 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

urate

family: purine · weak / non-responder · spike encodes: c17, c20



Interpretation. The spike drives components c7, c8, c22 most strongly (themes: protein, protein, protein). Its own encoding component(s) c17, c20 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is $\cos +0.07$ (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

histidine

family: amino_acid · weak / non-responder · spike encodes: c0, c15, c21



Interpretation. The spike drives components c7, c20, c3 most strongly (themes: protein, saccharide, sterol). Its own encoding component(s) c0, c15, c21 ARE among the strongest responders — an identity match. Pure-vs-serum consistency for this analyte is cos -0.13 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

acetoacetate

family: organic_acid · weak / non-responder · spike encodes: c5, c22



Interpretation. The spike drives components c3, c7, c20 most strongly (themes: sterol, protein, saccharide). Its own encoding component(s) c5, c22 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is $\cos +0.35$.

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

isoleucine

family: amino_acid · weak / non-responder · spike encodes: no atlas component



Interpretation. The spike drives components c7, c20, c3 most strongly (themes: protein, saccharide, sterol).
This analyte does not correspond to any single atlas component, so no identity match is possible.
Pure-vs-serum consistency for this analyte is cos -0.03 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

mannose

family: saccharide · weak / non-responder · spike encodes: c4, c5, c19



Interpretation. The spike drives components c7, c20, c3 most strongly (themes: protein, saccharide, sterol). Its own encoding component(s) c4, c5, c19 ARE among the strongest responders — an identity match. Pure-vs-serum consistency for this analyte is cos -0.08 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

arginine

family: amino_acid · weak / non-responder · spike encodes: c21

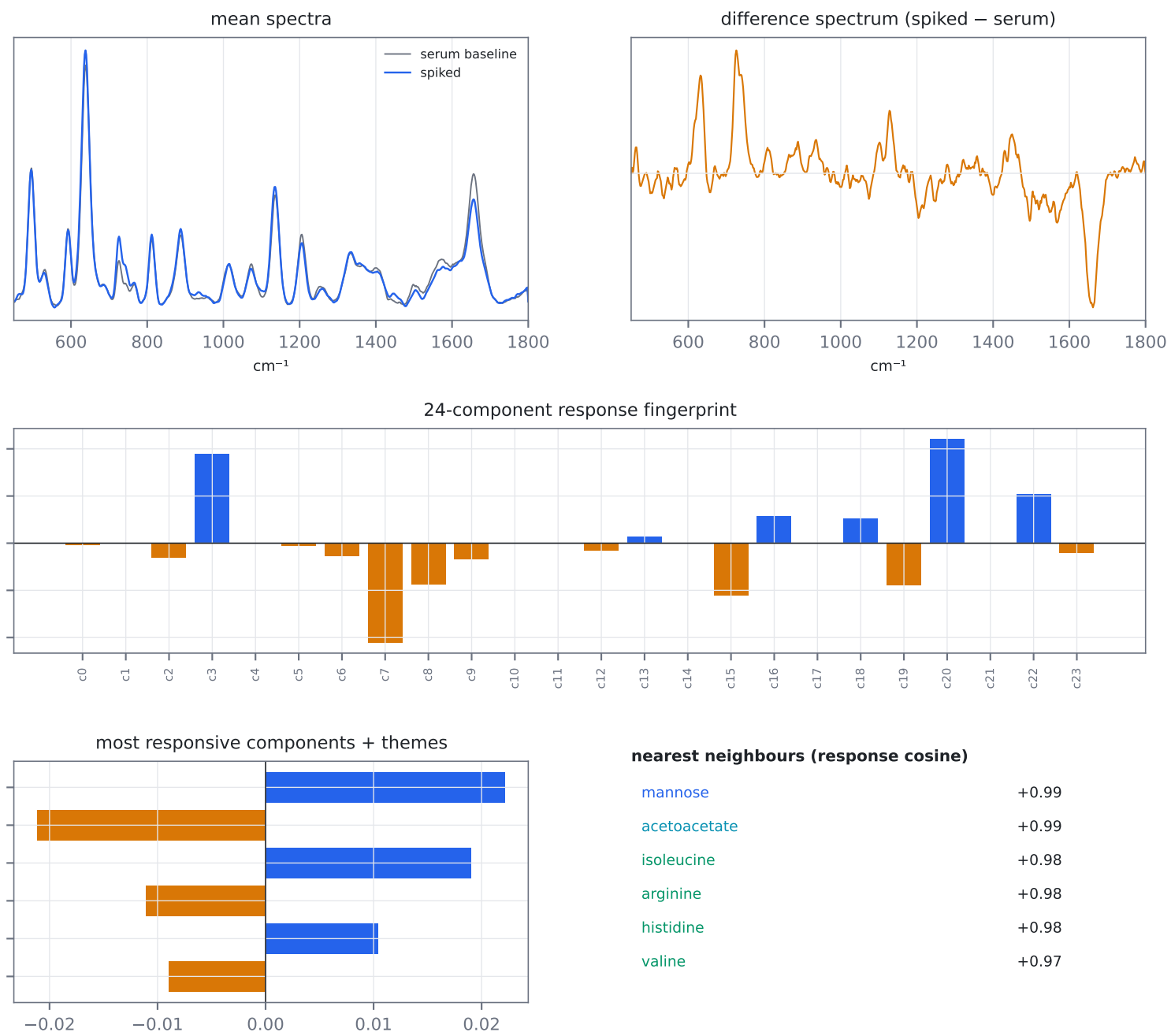


Interpretation. The spike drives components c7, c20, c3 most strongly (themes: protein, saccharide, sterol). Its own encoding component(s) c21 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is cos -0.12 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

stearate

family: lipid · weak / non-responder · spike encodes: no atlas component

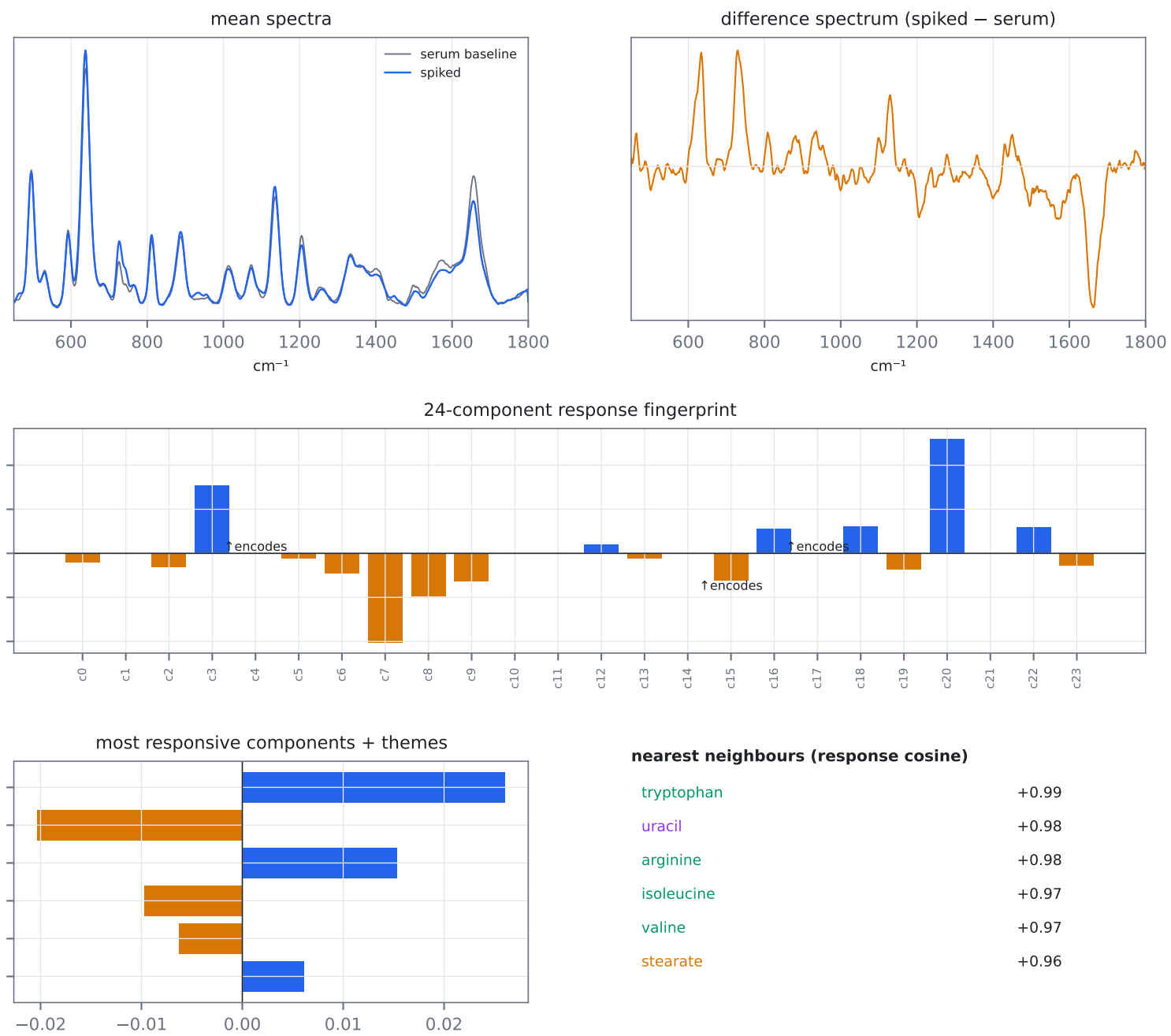


Interpretation. The spike drives components c20, c7, c3 most strongly (themes: saccharide, protein, sterol). This analyte does not correspond to any single atlas component, so no identity match is possible. Pure-vs-serum consistency for this analyte is cos +0.11 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

galactose

family: saccharide · weak / non-responder · spike encodes: c4, c15, c17



Interpretation. The spike drives components c20, c7, c3 most strongly (themes: saccharide, protein, sterol). Its own encoding component(s) c4, c15, c17 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is cos -0.01 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

proline

family: amino_acid · weak / non-responder · spike encodes: c9, c10

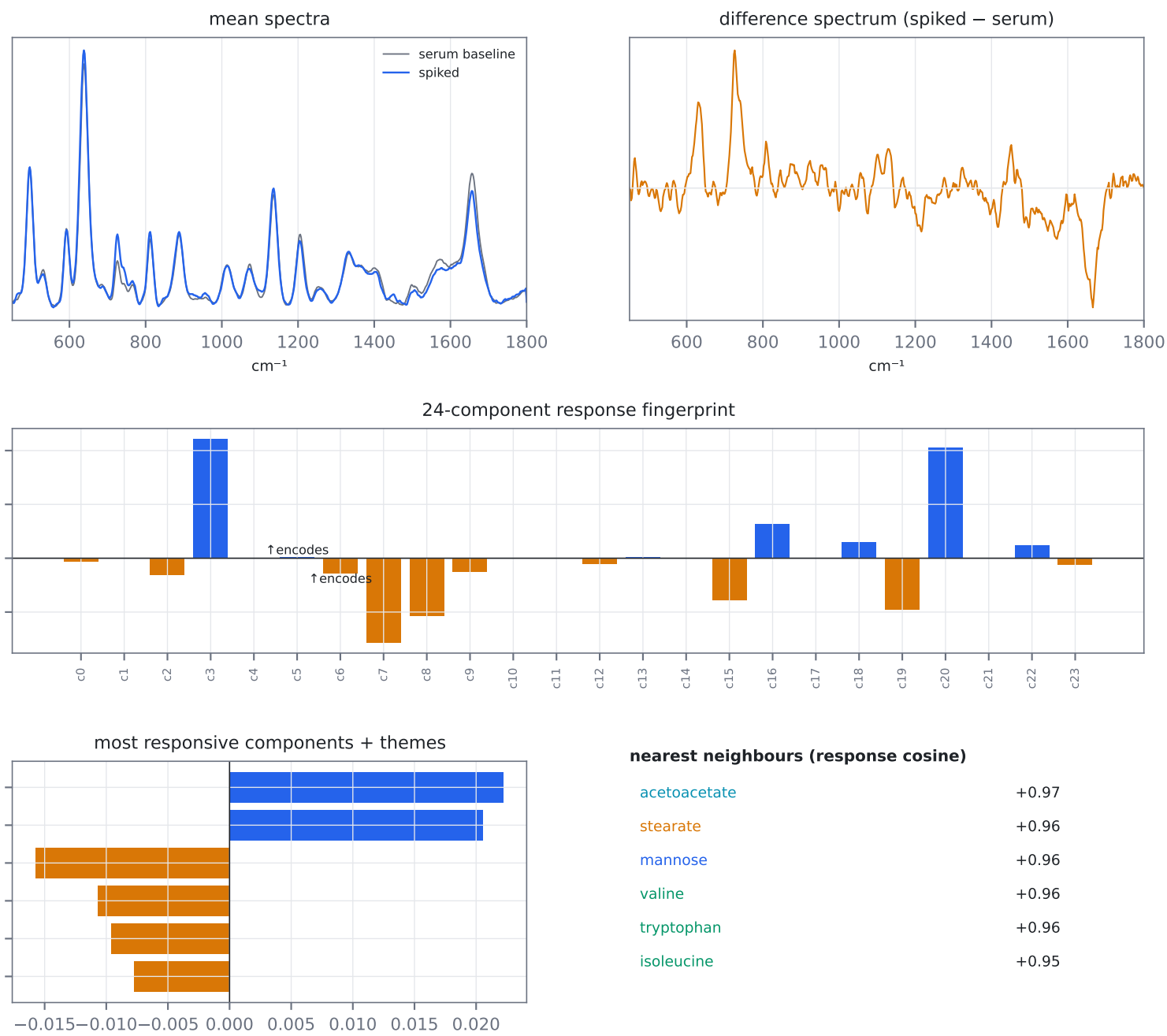


Interpretation. The spike drives components c7, c20, c3 most strongly (themes: protein, saccharide, sterol). Its own encoding component(s) c9, c10 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is cos +0.17 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

phenylalanine

family: amino_acid · weak / non-responder · spike encodes: c5, c6



Interpretation. The spike drives components c3, c20, c7 most strongly (themes: sterol, saccharide, protein). Its own encoding component(s) c5, c6 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is cos +0.06 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

tryptophan

family: amino_acid · weak / non-responder · spike encodes: c0, c6, c22



Interpretation. The spike drives components c20, c7, c3 most strongly (themes: saccharide, protein, sterol). Its own encoding component(s) c0, c6, c22 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is cos -0.19 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

uracil

family: pyrimidine · weak / non-responder · spike encodes: c17



Interpretation. The spike drives components c20, c7, c3 most strongly (themes: saccharide, protein, sterol). Its own encoding component(s) c17 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is cos +0.12 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

valine

family: amino_acid · weak / non-responder · spike encodes: c12



Interpretation. The spike drives components c20, c7, c3 most strongly (themes: saccharide, protein, sterol). Its own encoding component(s) c12 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is $\cos +0.03$ (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

coenzyme a

family: cofactor · weak / non-responder · spike encodes: c3



Interpretation. The spike drives components c20, c3, c7 most strongly (themes: saccharide, sterol, protein). Its own encoding component(s) c3 ARE among the strongest responders — an identity match. Pure-vs-serum consistency for this analyte is cos -0.23 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

tyrosine

family: amino_acid · weak / non-responder · spike encodes: c5, c15



Interpretation. The spike drives components c20, c3, c15 most strongly (themes: saccharide, sterol, purine). Its own encoding component(s) c5, c15 ARE among the strongest responders — an identity match. Pure-vs-serum consistency for this analyte is cos +0.01 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

lactate

family: organic_acid · weak / non-responder · spike encodes: c18

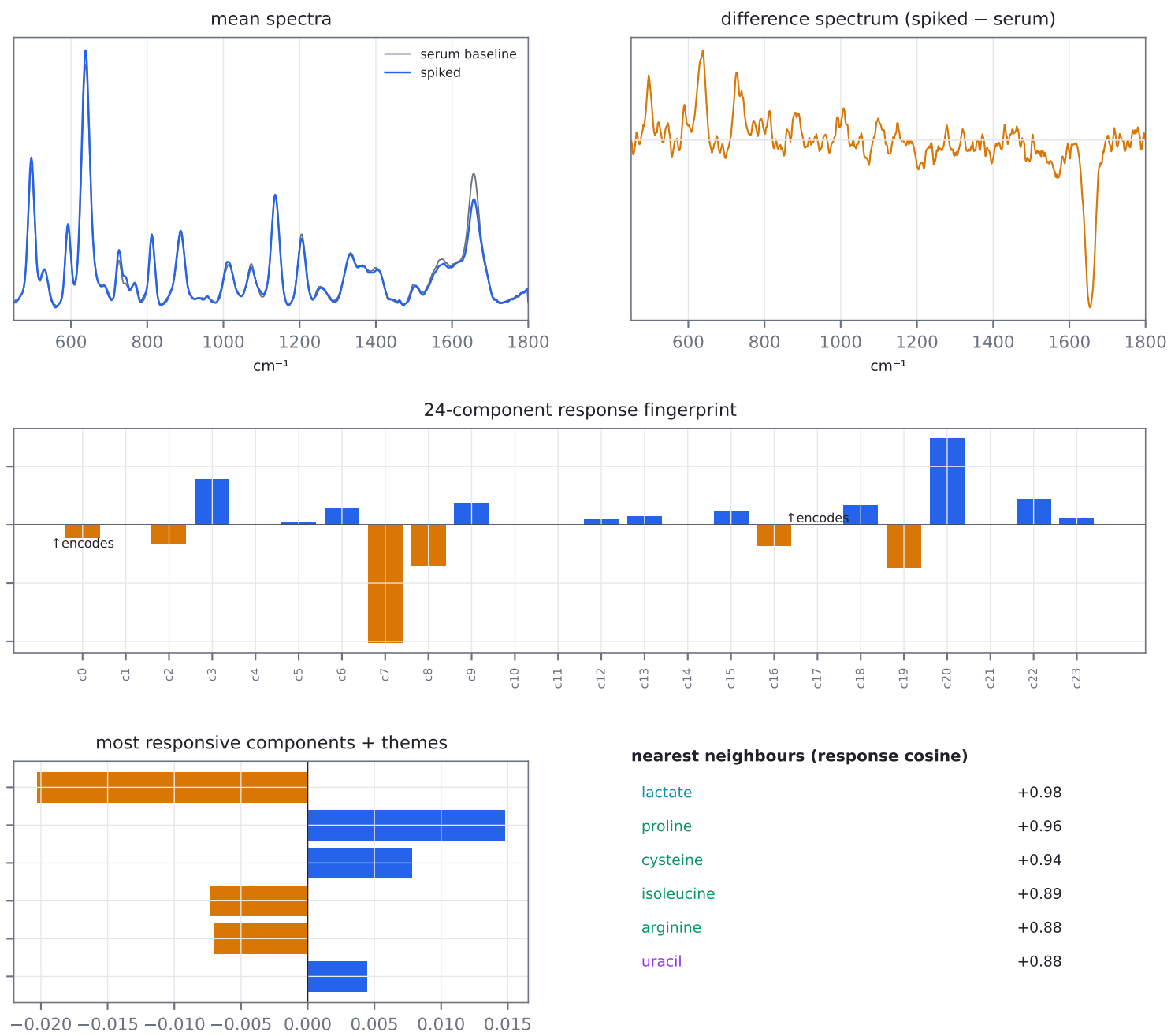


Interpretation. The spike drives components c7, c20, c3 most strongly (themes: protein, saccharide, sterol). Its own encoding component(s) c18 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is cos -0.31 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

riboflavin

family: cofactor · weak / non-responder · spike encodes: c0, c17

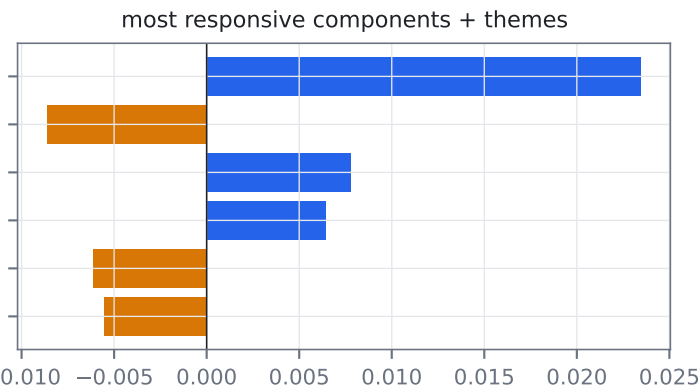
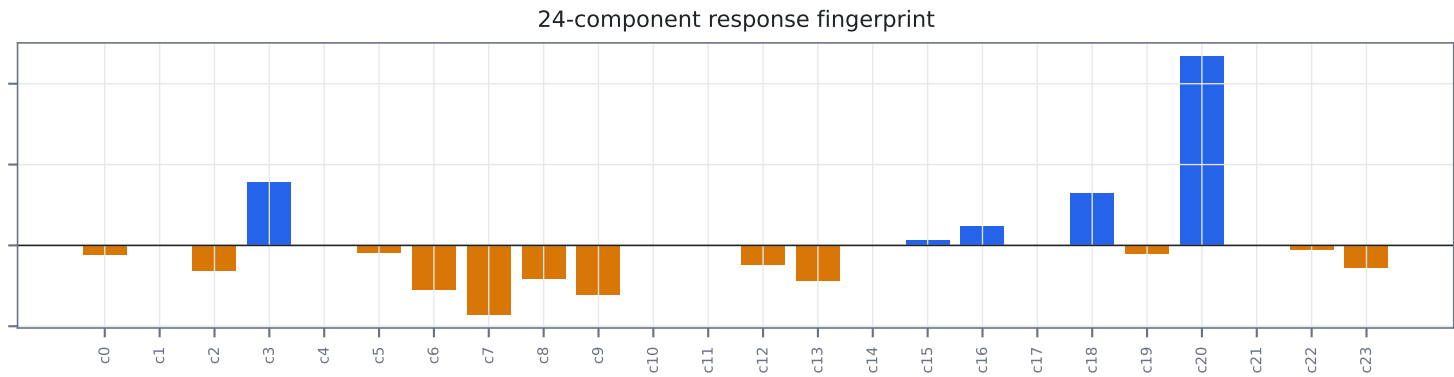
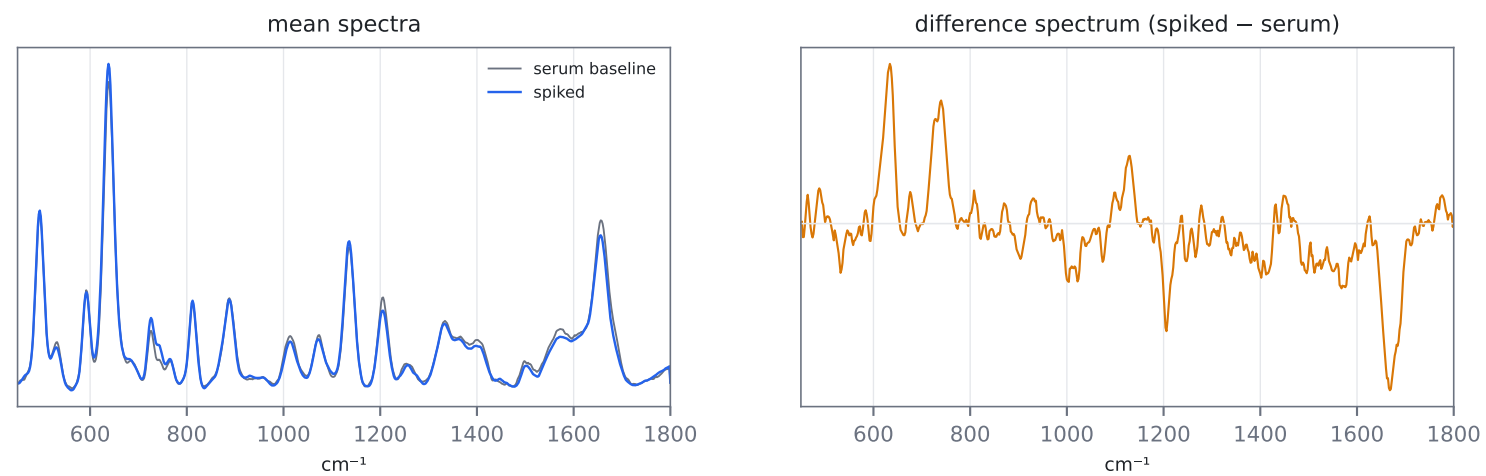


Interpretation. The spike drives components c7, c20, c3 most strongly (themes: protein, saccharide, sterol). Its own encoding component(s) c0, c17 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is $\cos -0.04$ (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

glycogen

family: polysaccharide · weak / non-responder · spike encodes: no atlas component



nearest neighbours (response cosine)

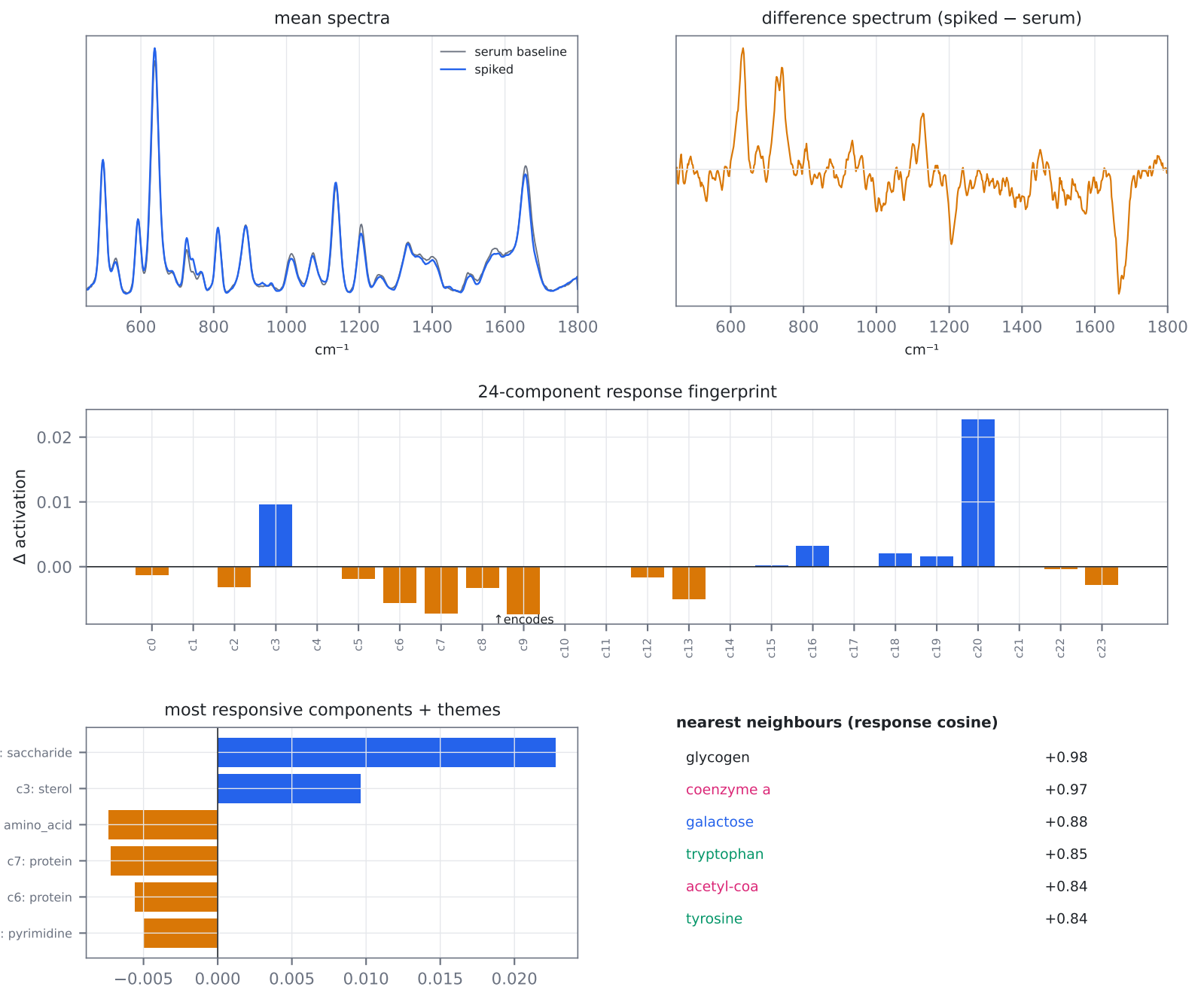
glutathione	+0.98
coenzyme a	+0.97
galactose	+0.90
tryptophan	+0.86
cysteine	+0.86
uracil	+0.86

Interpretation. The spike drives components c20, c7, c3 most strongly (themes: saccharide, protein, sterol). This analyte does not correspond to any single atlas component, so no identity match is possible. Pure-vs-serum consistency for this analyte is cos +0.06 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

glutathione

family: cofactor · weak / non-responder · spike encodes: c9

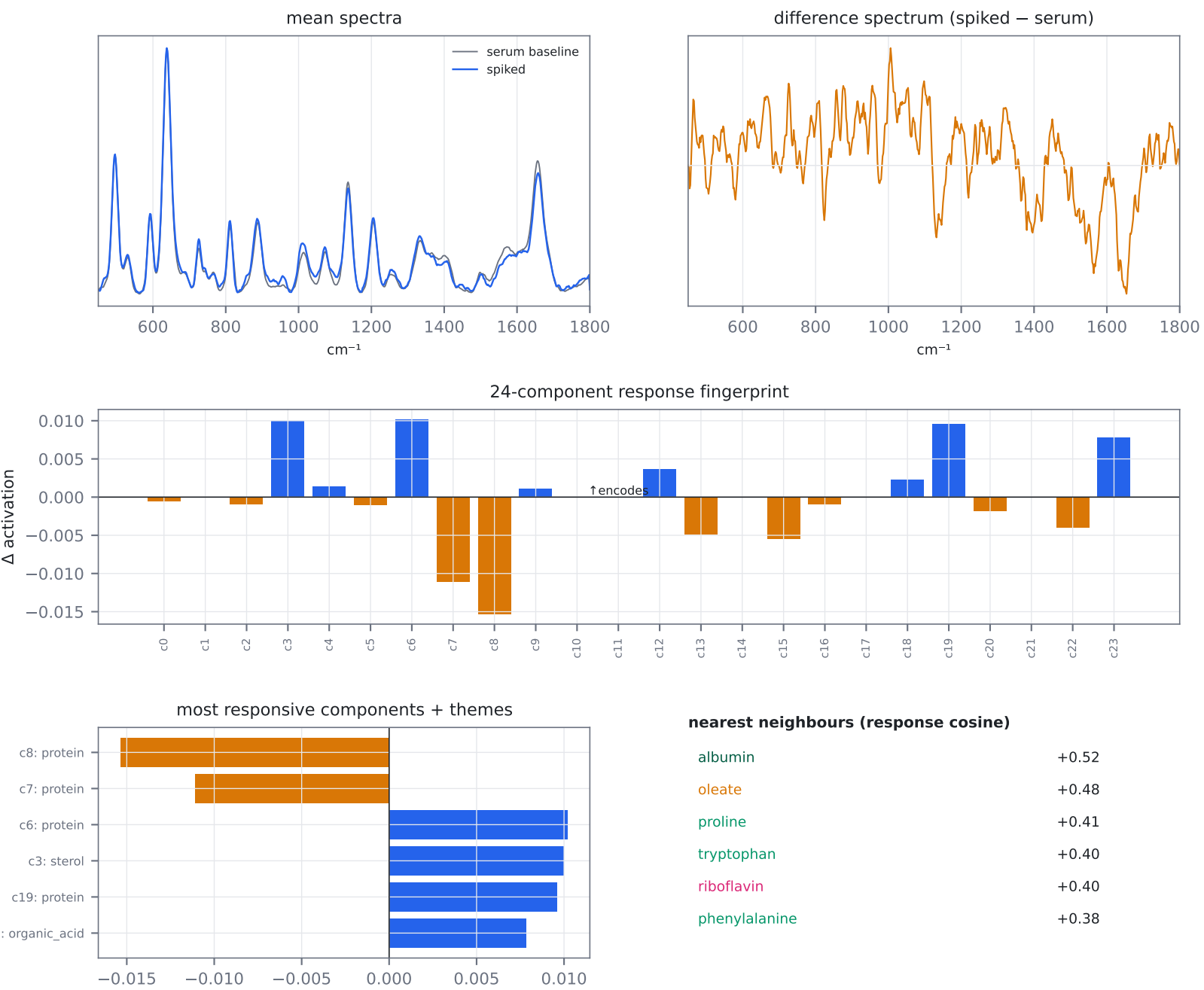


Interpretation. The spike drives components c20, c3, c9 most strongly (themes: saccharide, sterol, amino_acid). Its own encoding component(s) c9 ARE among the strongest responders — an identity match. Pure-vs-serum consistency for this analyte is cos -0.10 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

serine

family: amino_acid · weak / non-responder · spike encodes: c11



Interpretation. The spike drives components c8, c7, c6 most strongly (themes: protein, protein, protein). Its own encoding component(s) c11 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is cos +0.02 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

fructose

family: saccharide · weak / non-responder · spike encodes: c4, c20

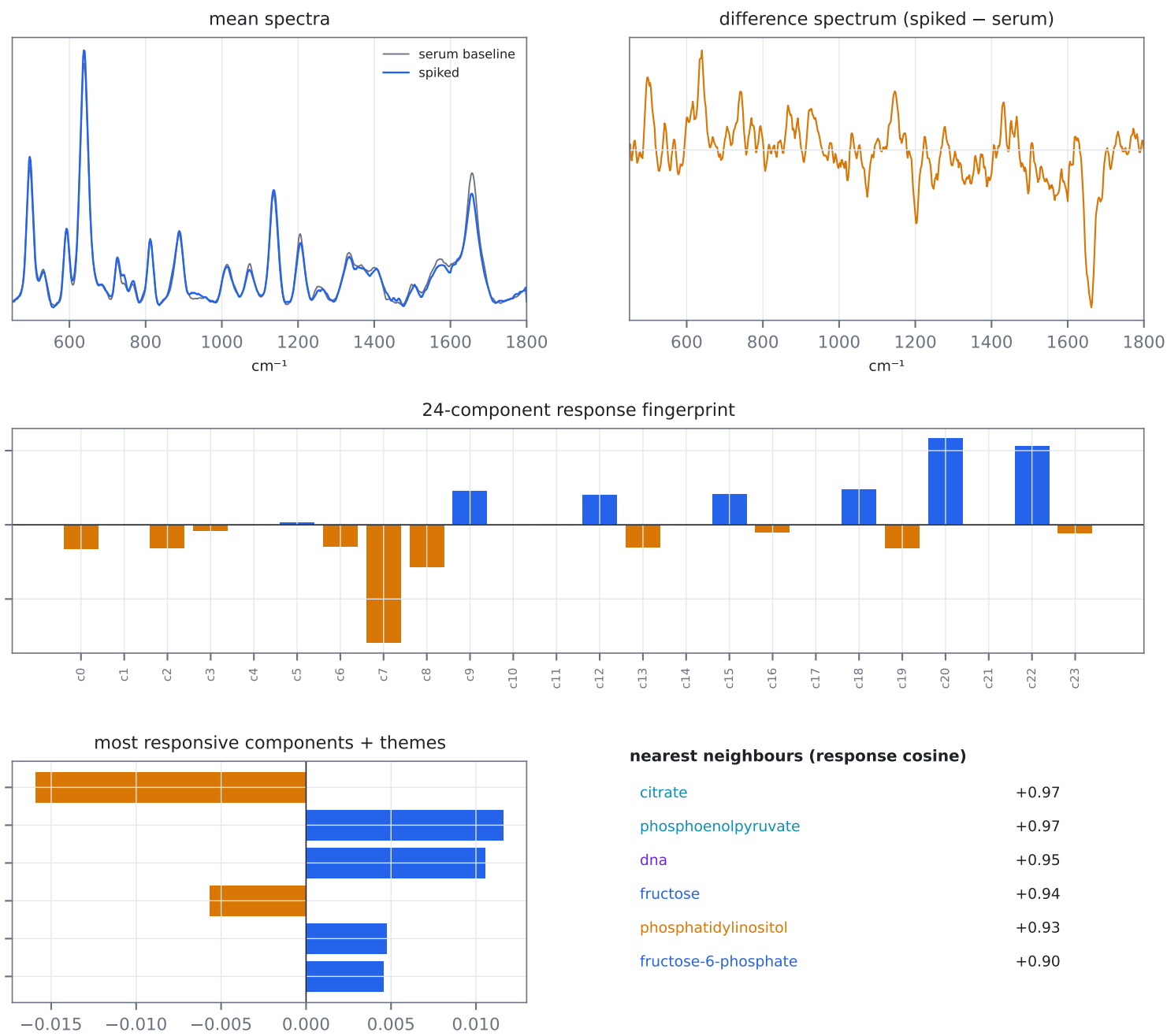


Interpretation. The spike drives components c7, c20, c22 most strongly (themes: protein, saccharide, protein). Its own encoding component(s) c4, c20 ARE among the strongest responders — an identity match. Pure-vs-serum consistency for this analyte is cos +0.12 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

rna

family: nucleic_acid · weak / non-responder · spike encodes: no atlas component



Interpretation. The spike drives components c7, c20, c22 most strongly (themes: protein, saccharide, protein). This analyte does not correspond to any single atlas component, so no identity match is possible. Pure-vs-serum consistency for this analyte is cos -0.25 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

cysteine

family: amino_acid · weak / non-responder · spike encodes: c15, c19, c20



Interpretation. The spike drives components c20, c7, c3 most strongly (themes: saccharide, protein, sterol). Its own encoding component(s) c15, c19, c20 ARE among the strongest responders — an identity match. Pure-vs-serum consistency for this analyte is cos -0.02 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

dna

family: nucleic_acid · weak / non-responder · spike encodes: no atlas component

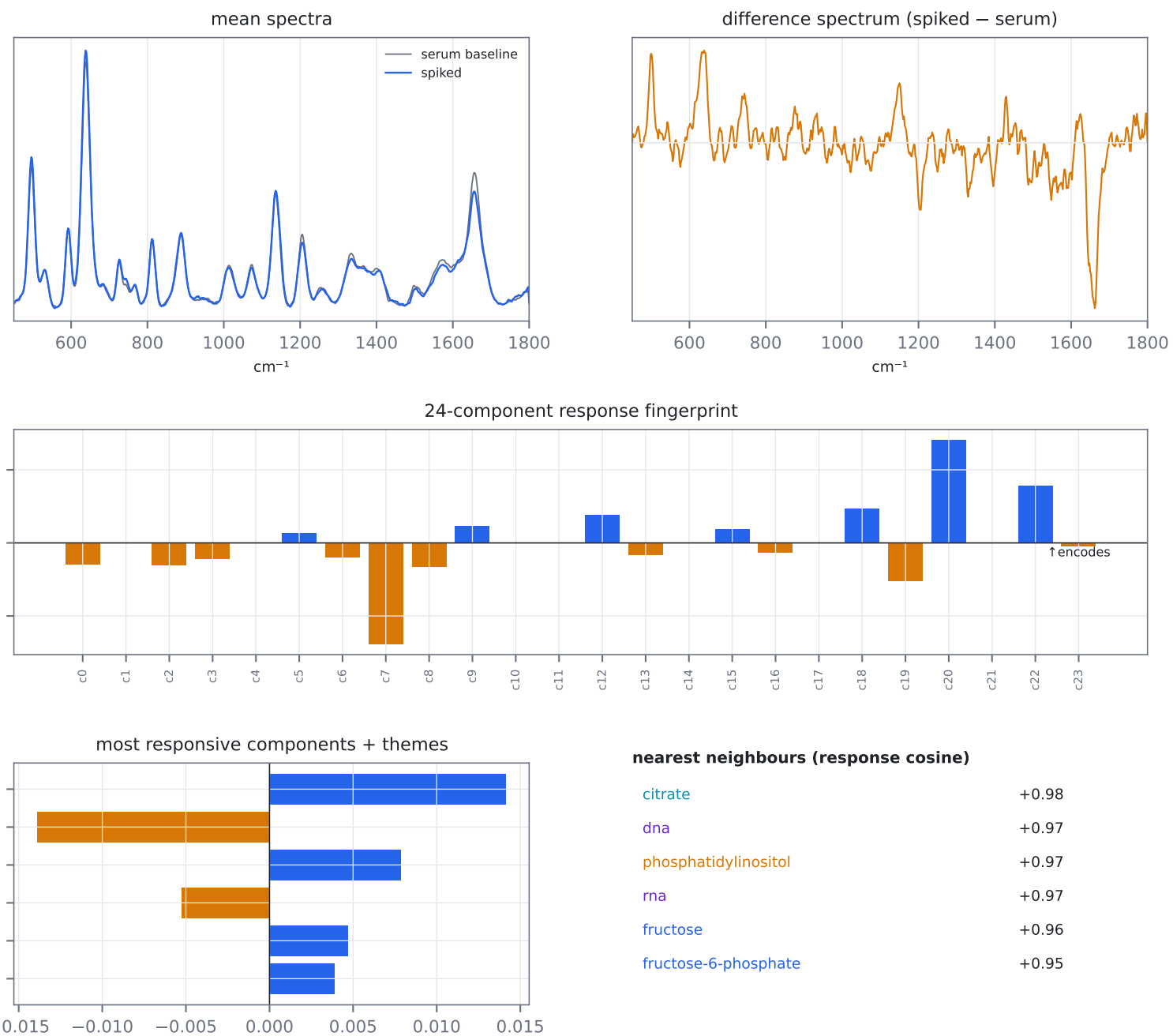


Interpretation. The spike drives components c7, c20, c22 most strongly (themes: protein, saccharide, protein). This analyte does not correspond to any single atlas component, so no identity match is possible. Pure-vs-serum consistency for this analyte is $\cos -0.05$ (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

phosphoenolpyruvate

family: organic_acid · weak / non-responder · spike encodes: c23

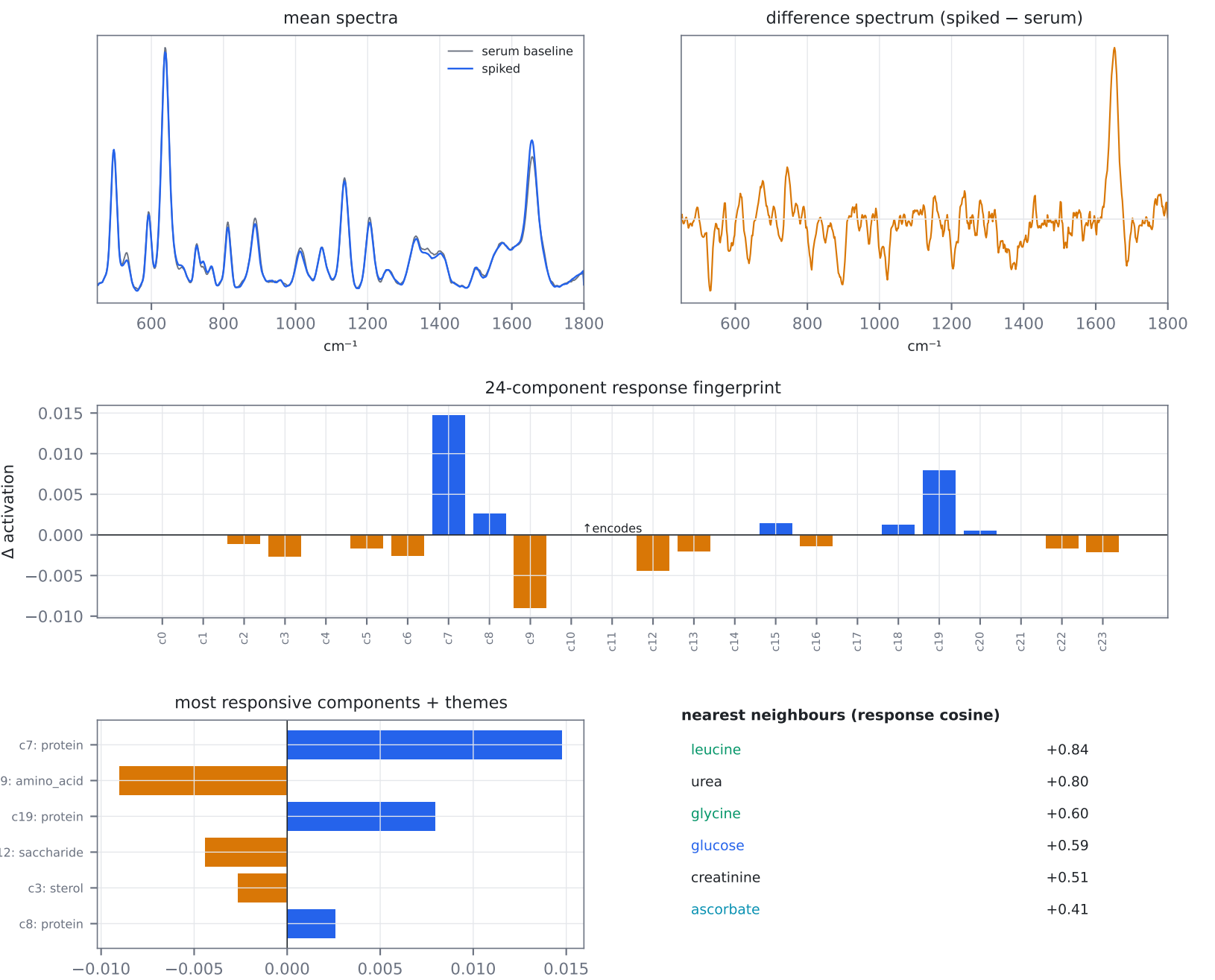


Interpretation. The spike drives components c20, c7, c22 most strongly (themes: saccharide, protein, protein). Its own encoding component(s) c23 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is cos -0.16 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

glycerol

family: polyol · weak / non-responder · spike encodes: c11

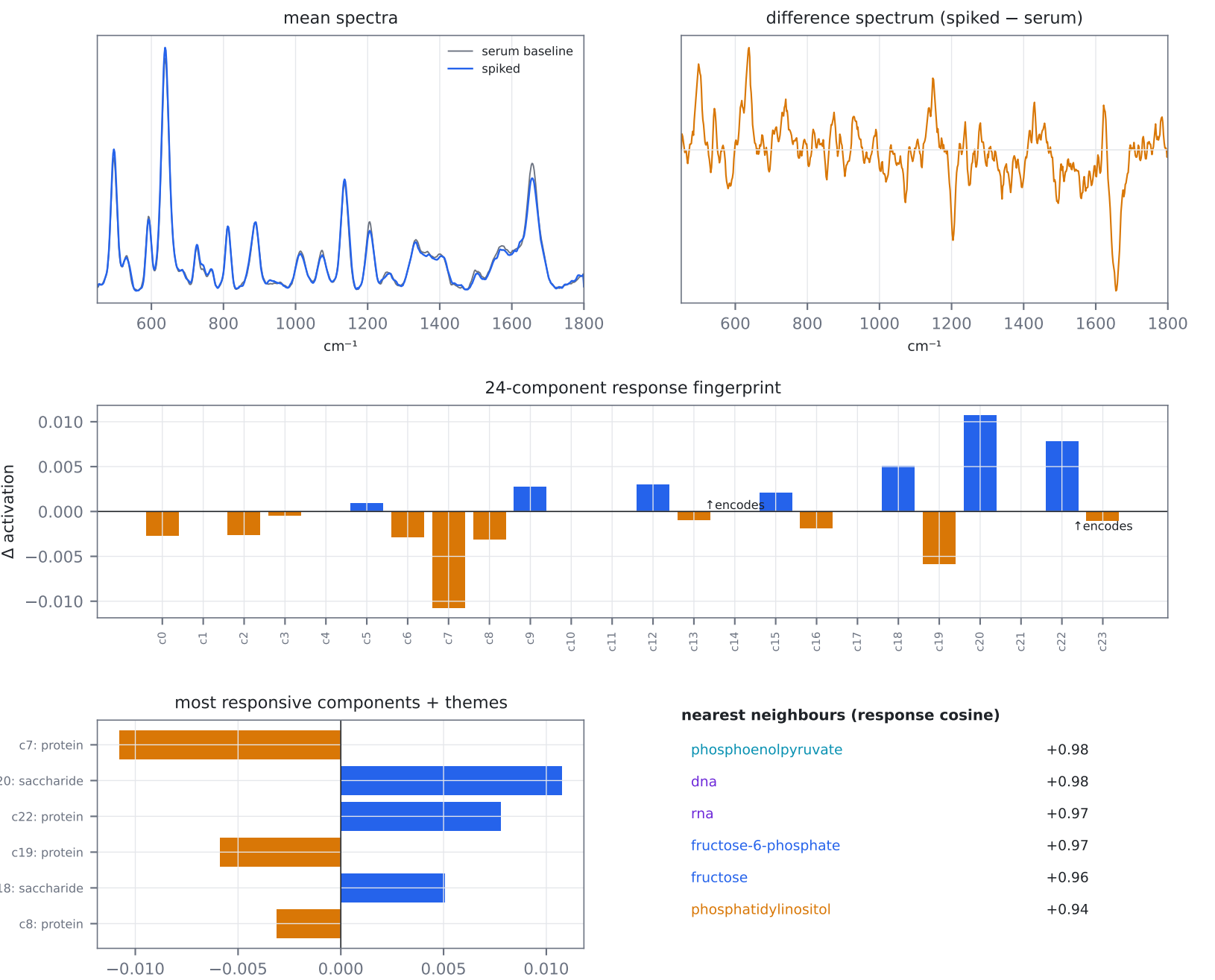


Interpretation. The spike drives components c7, c9, c19 most strongly (themes: protein, amino_acid, protein). Its own encoding component(s) c11 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is cos +0.26 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

citrate

family: organic_acid · weak / non-responder · spike encodes: c14, c23

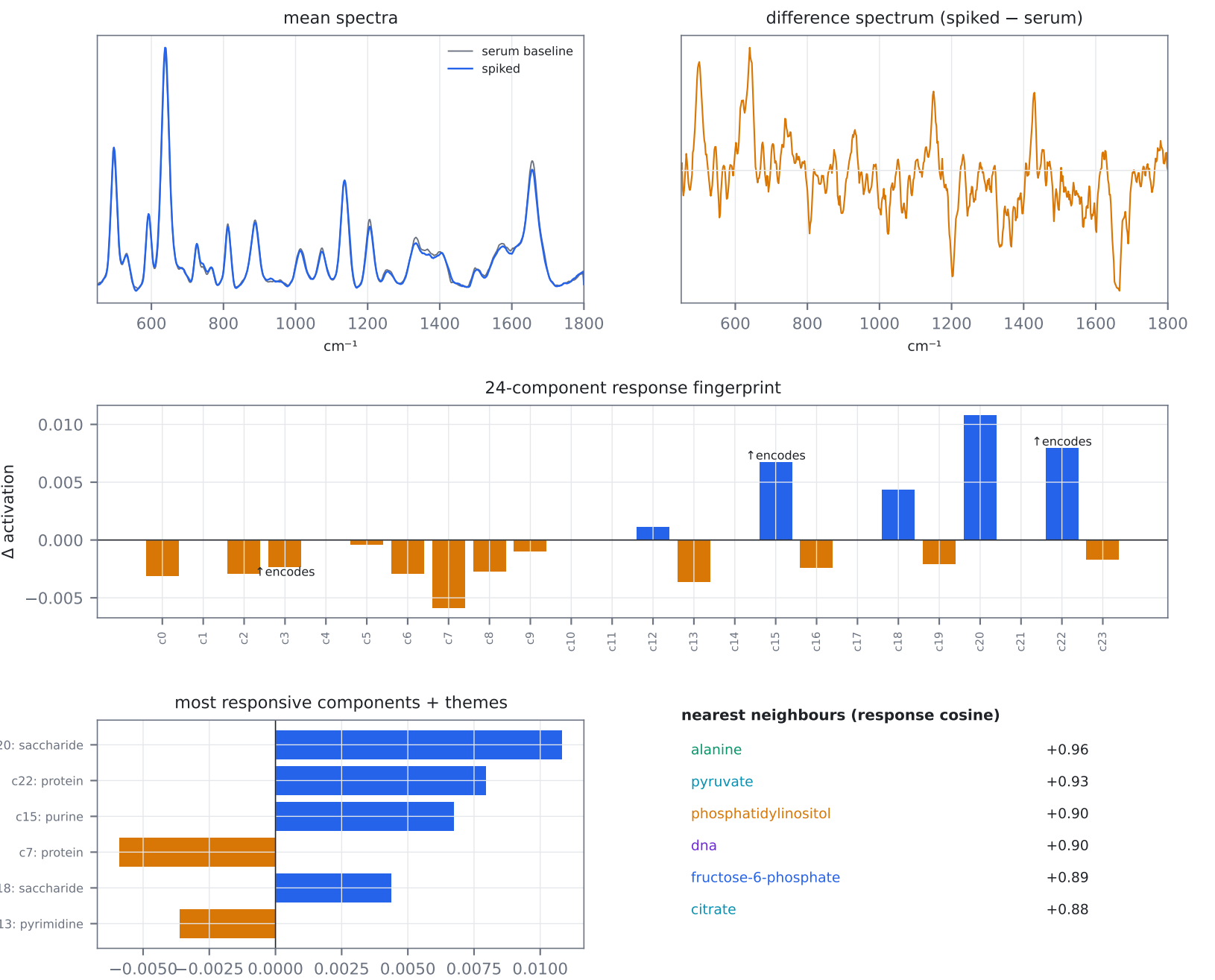


Interpretation. The spike drives components c7, c20, c22 most strongly (themes: protein, saccharide, protein). Its own encoding component(s) c14, c23 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is $\cos +0.04$ (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

methionine

family: amino_acid · weak / non-responder · spike encodes: c3, c15, c22

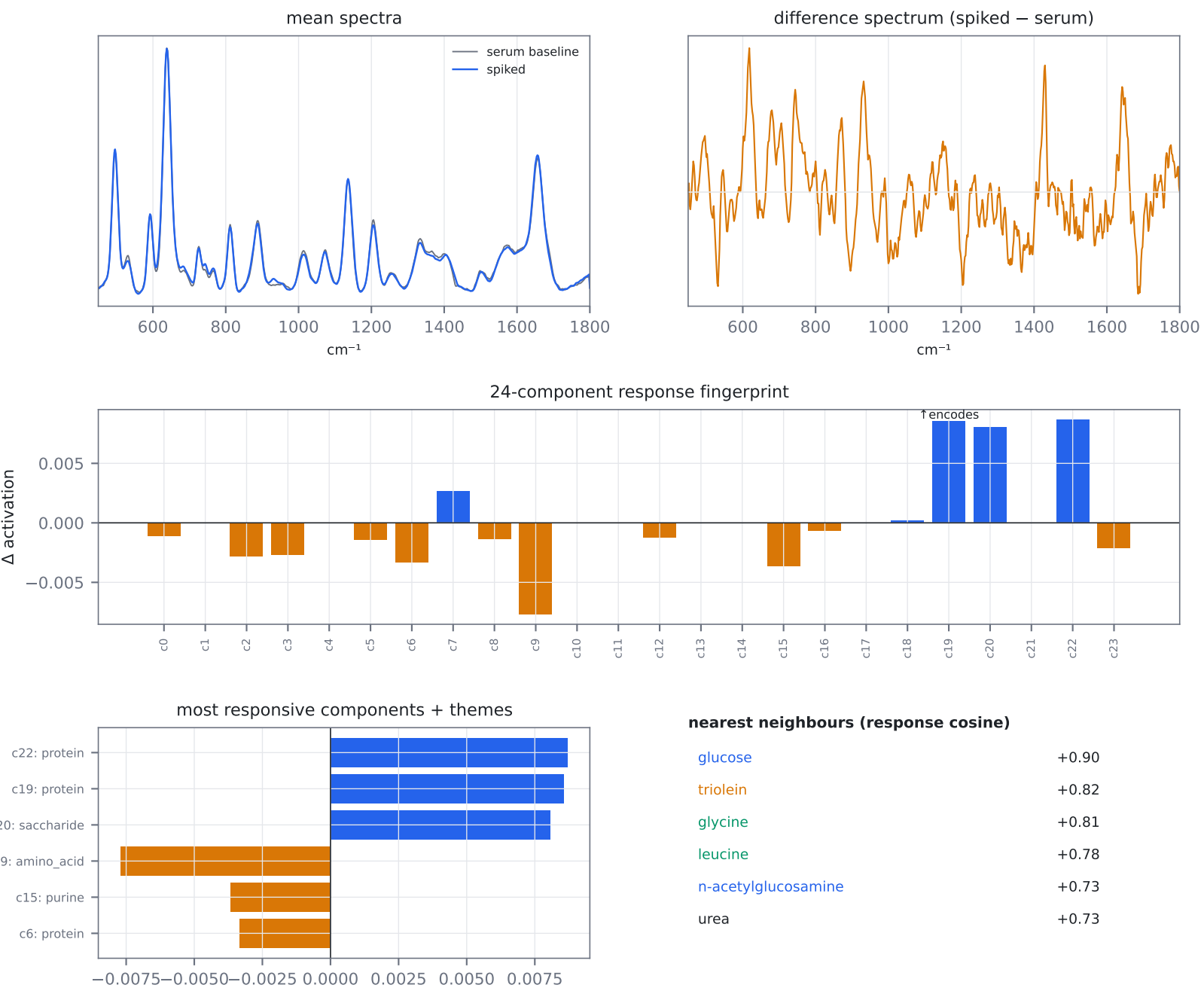


Interpretation. The spike drives components c20, c22, c15 most strongly (themes: saccharide, protein, purine). Its own encoding component(s) c3, c15, c22 ARE among the strongest responders — an identity match. Pure-vs-serum consistency for this analyte is cos -0.23 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

creatinine

family: small_nitrogenous · RESPONDER · spike encodes: c19

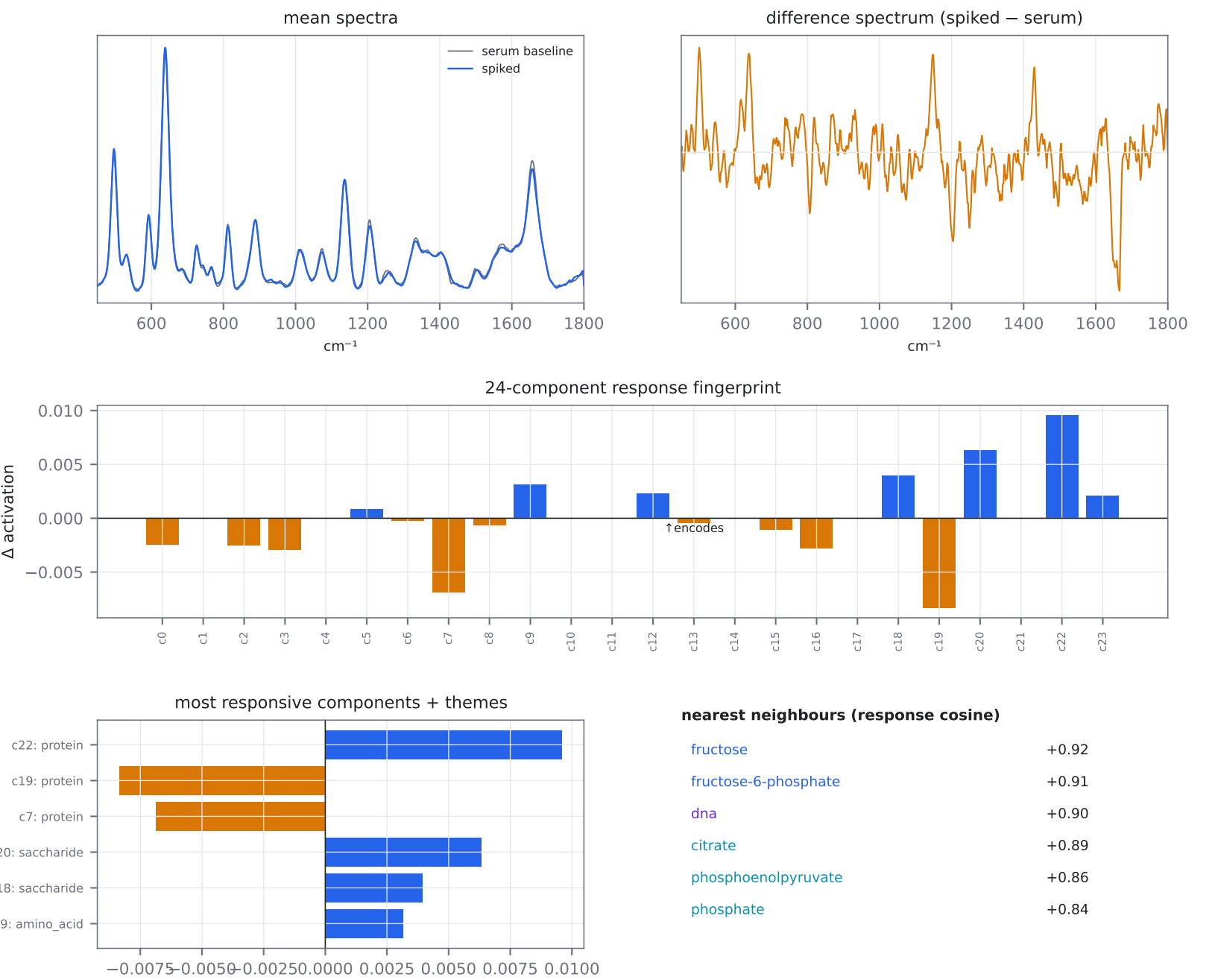


Interpretation. The spike drives components c22, c19, c20 most strongly (themes: protein, protein, saccharide). Its own encoding component(s) c19 ARE among the strongest responders — an identity match. Pure-vs-serum consistency for this analyte is $\cos +0.58$.

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

thymine

family: pyrimidine · weak / non-responder · spike encodes: c13

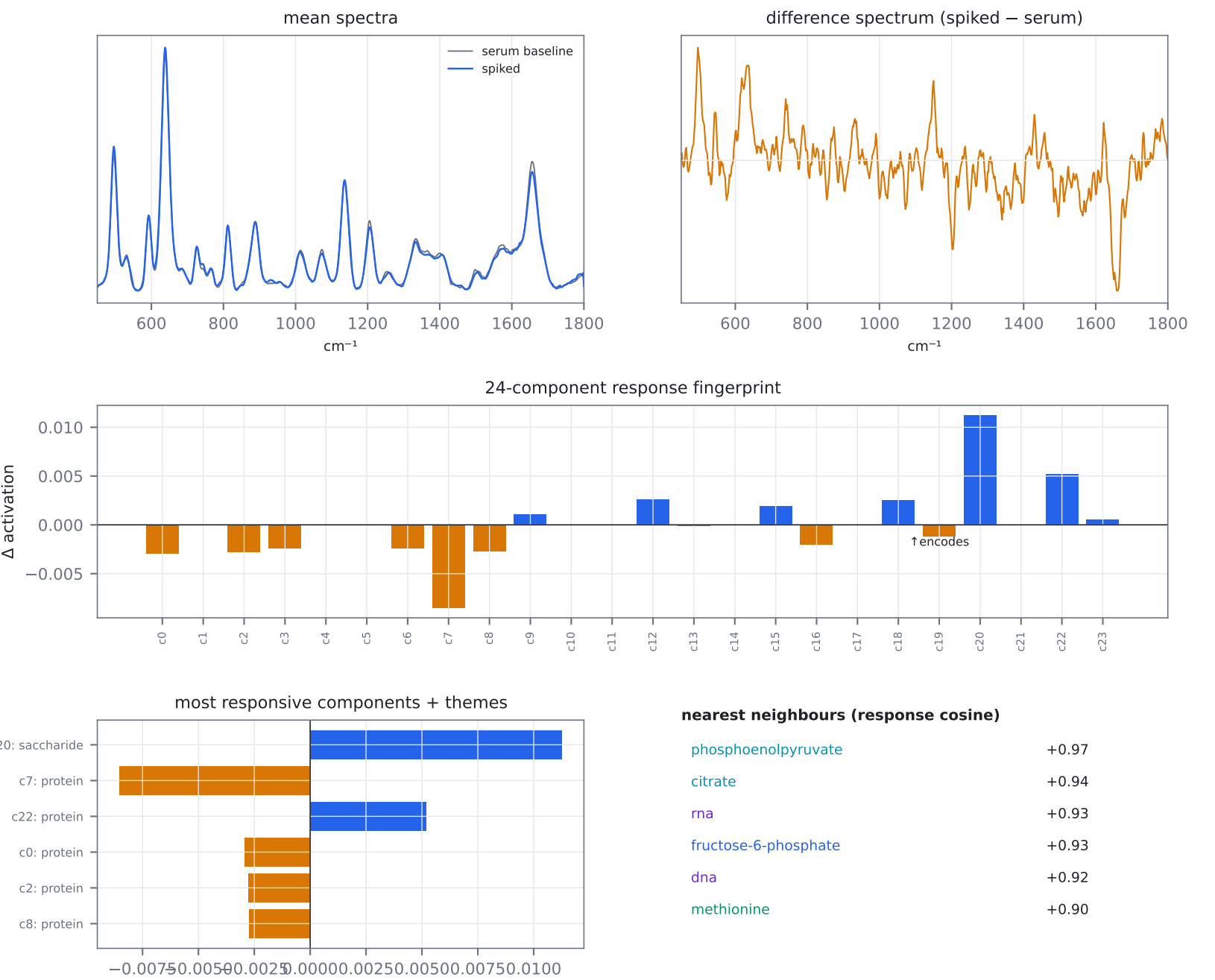


Interpretation. The spike drives components c22, c19, c7 most strongly (themes: protein, protein, protein). Its own encoding component(s) c13 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is $\cos -0.07$ (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

phosphatidylinositol

family: lipid · weak / non-responder · spike encodes: c19

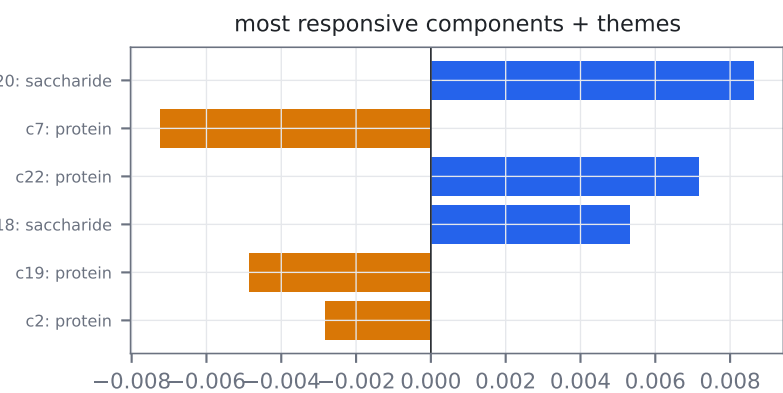
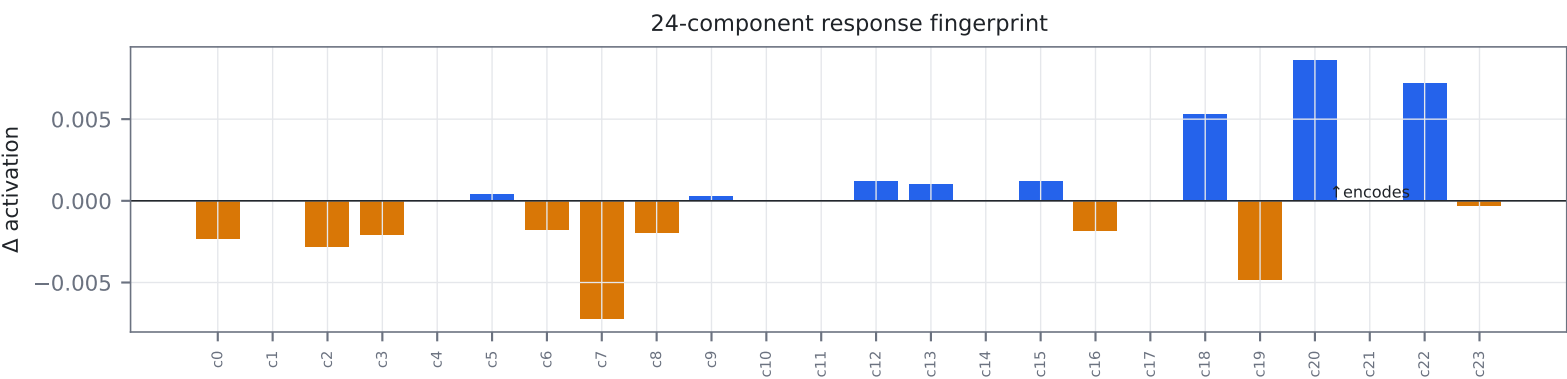
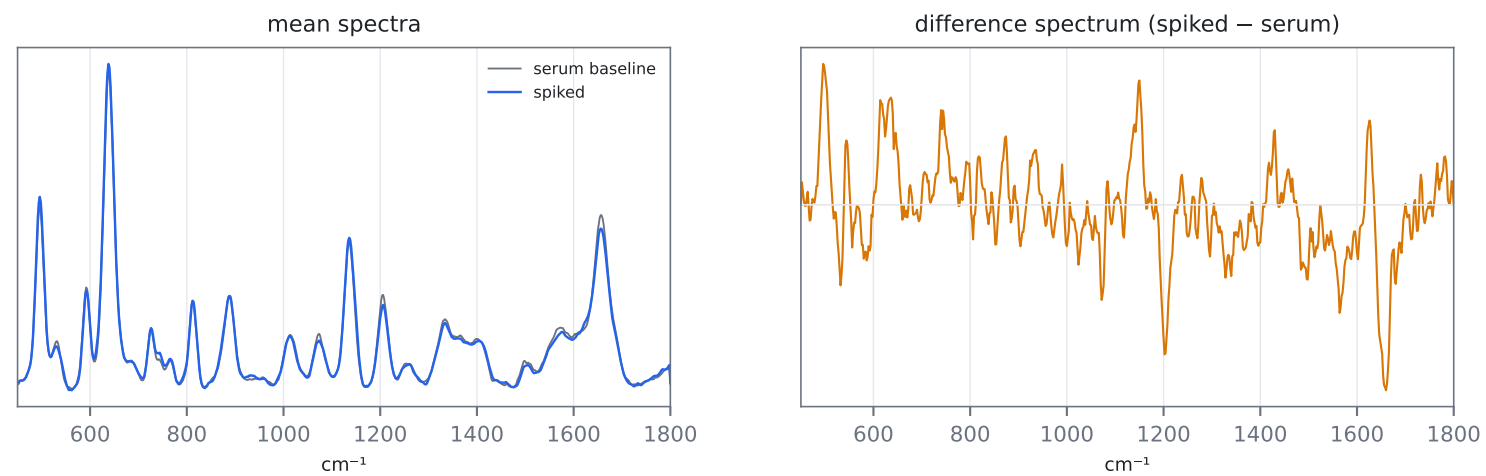


Interpretation. The spike drives components c20, c7, c22 most strongly (themes: saccharide, protein, protein). Its own encoding component(s) c19 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is cos +0.06 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

fructose-6-phosphate

family: saccharide · weak / non-responder · spike encodes: c21



nearest neighbours (response cosine)

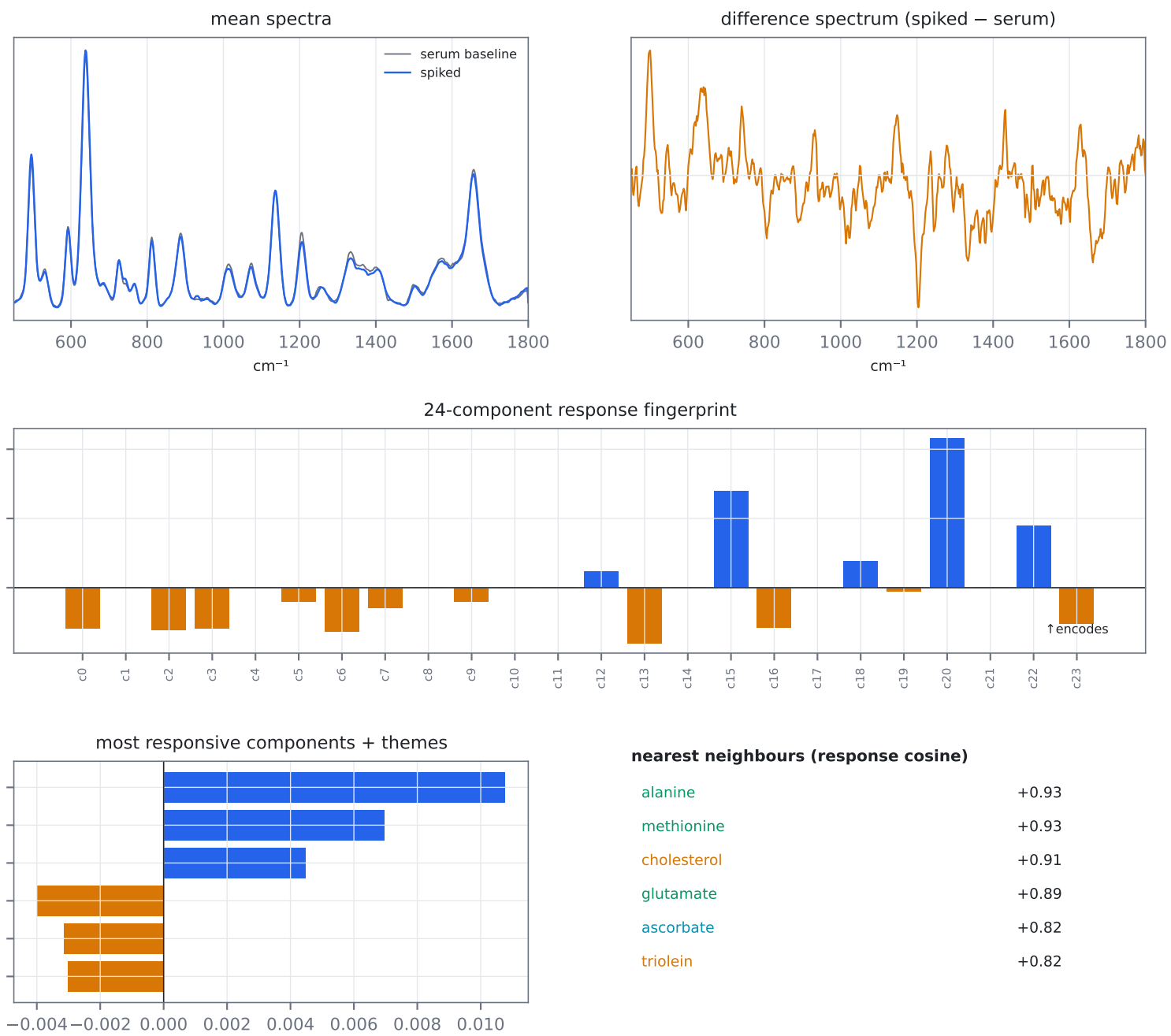
dna	+0.97
citrate	+0.97
phosphoenolpyruvate	+0.95
fructose	+0.93
phosphatidylinositol	+0.93
thymine	+0.91

Interpretation. The spike drives components c20, c7, c22 most strongly (themes: saccharide, protein, protein). Its own encoding component(s) c21 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is $\cos -0.27$ (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

pyruvate

family: organic_acid · weak / non-responder · spike encodes: c23

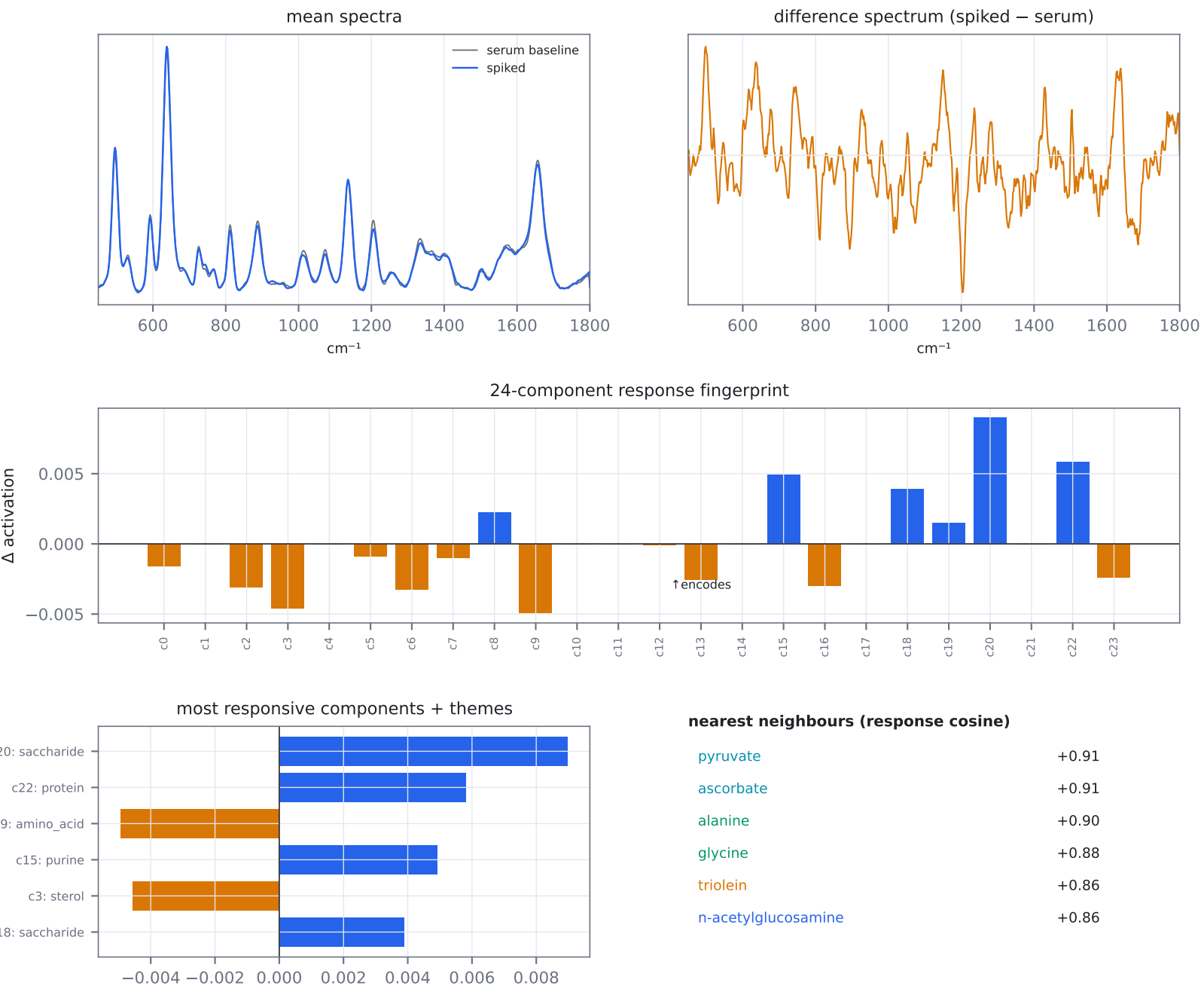


Interpretation. The spike drives components c20, c15, c22 most strongly (themes: saccharide, purine, protein). Its own encoding component(s) c23 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is $\cos +0.05$ (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

cholesterol

family: lipid · weak / non-responder · spike encodes: c13

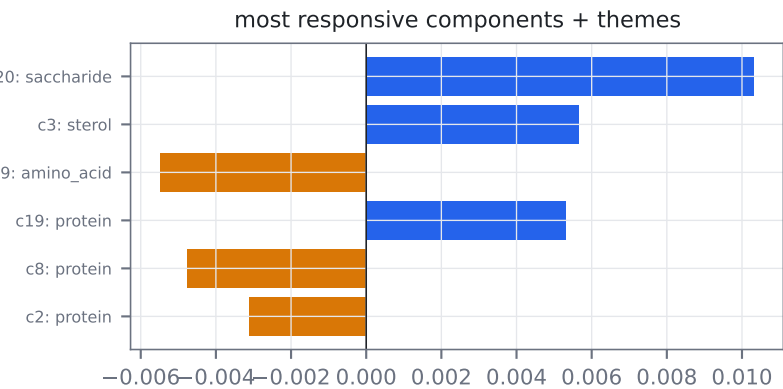
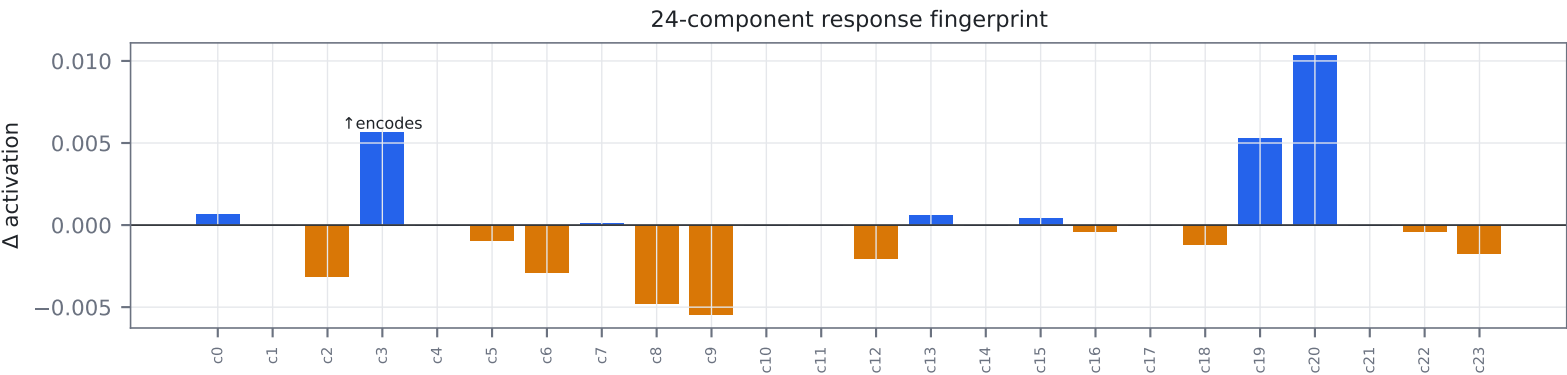
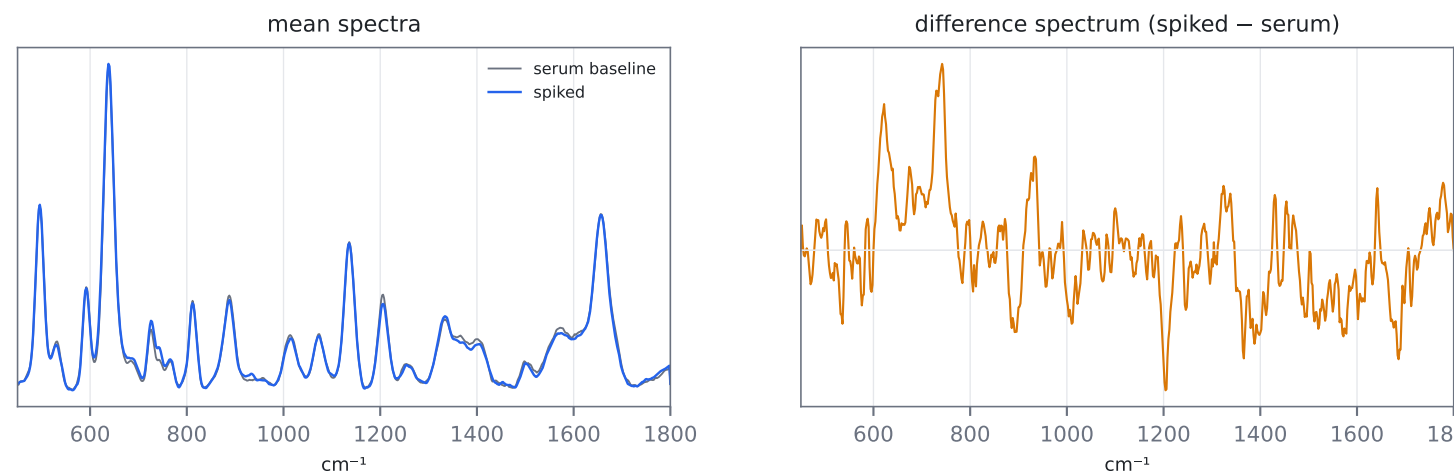


Interpretation. The spike drives components c20, c22, c9 most strongly (themes: saccharide, protein, amino_acid). Its own encoding component(s) c13 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is cos -0.16 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

acetyl-coa

family: cofactor · weak / non-responder · spike encodes: c3



nearest neighbours (response cosine)

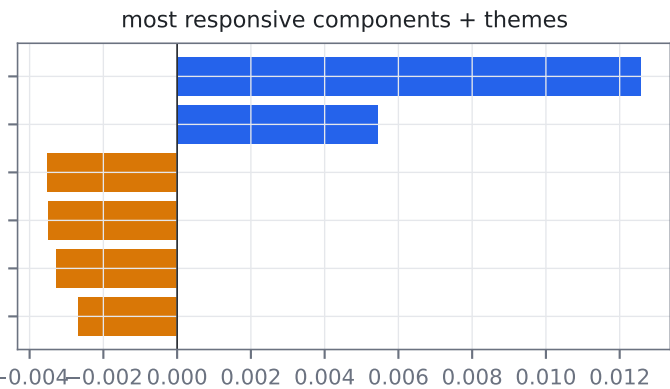
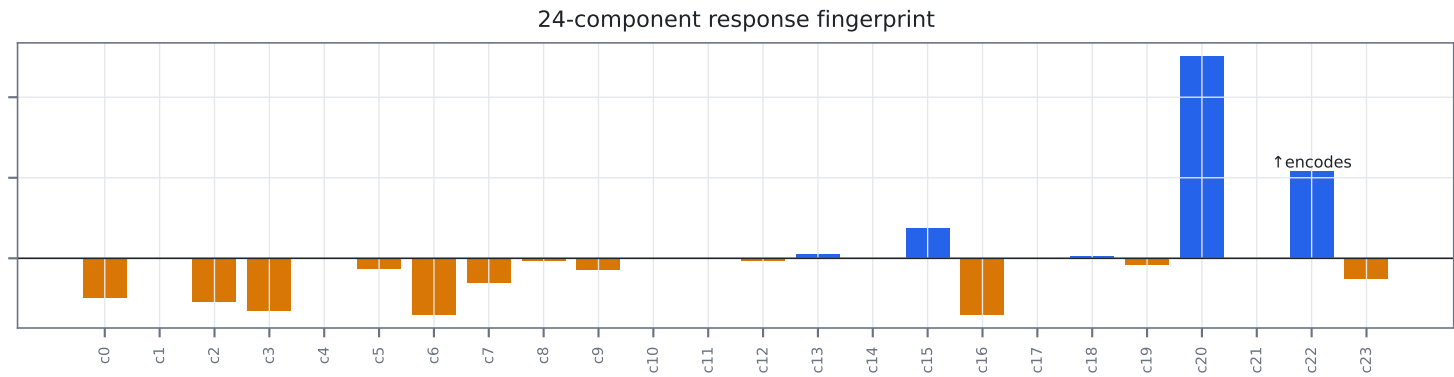
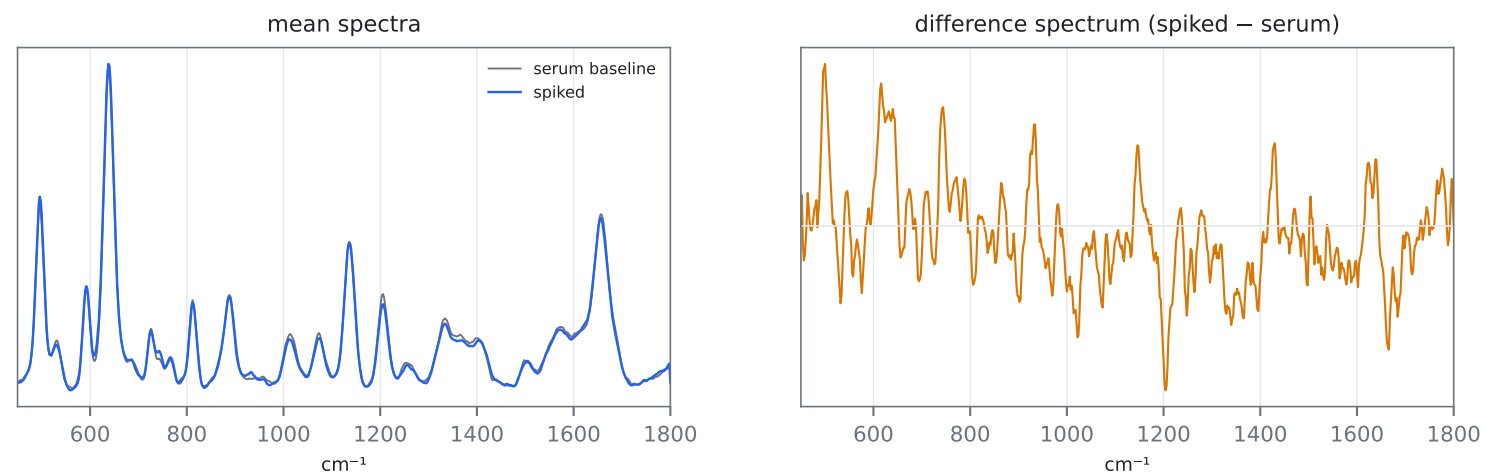
glutathione	+0.84
coenzyme a	+0.78
glycogen	+0.77
tyrosine	+0.68
galactose	+0.66
triolein	+0.65

Interpretation. The spike drives components c20, c3, c9 most strongly (themes: saccharide, sterol, amino_acid). Its own encoding component(s) c3 ARE among the strongest responders — an identity match. Pure-vs-serum consistency for this analyte is cos -0.18 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

glutamate

family: amino_acid · weak / non-responder · spike encodes: c22



nearest neighbours (response cosine)

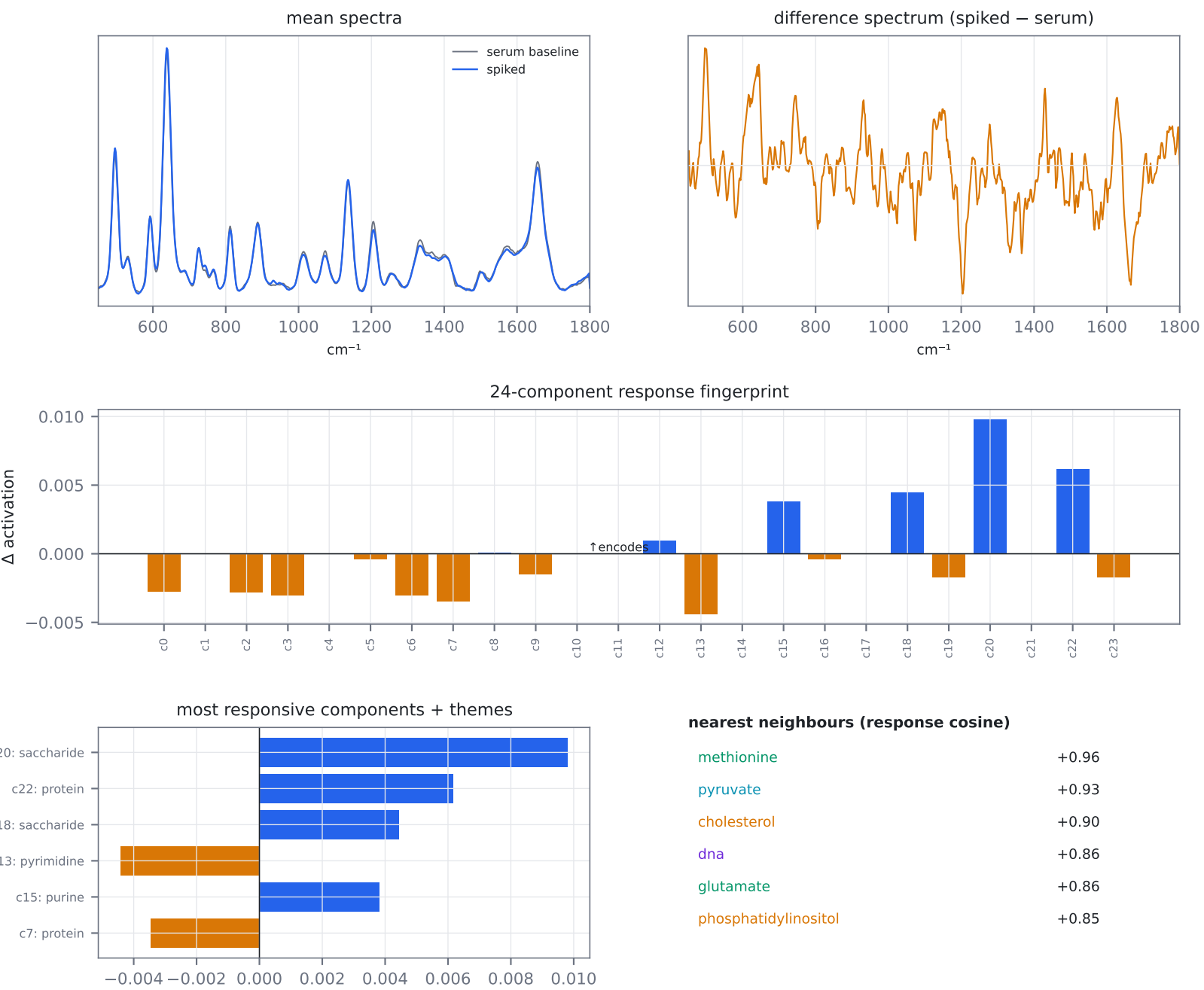
n-acetylglucosamine	+0.93
triolein	+0.91
pyruvate	+0.89
ascorbate	+0.86
alanine	+0.86
cholesterol	+0.85

Interpretation. The spike drives components c20, c22, c16 most strongly (themes: saccharide, protein, triglyceride). Its own encoding component(s) c22 ARE among the strongest responders — an identity match. Pure-vs-serum consistency for this analyte is cos -0.28 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

alanine

family: amino_acid · weak / non-responder · spike encodes: c11

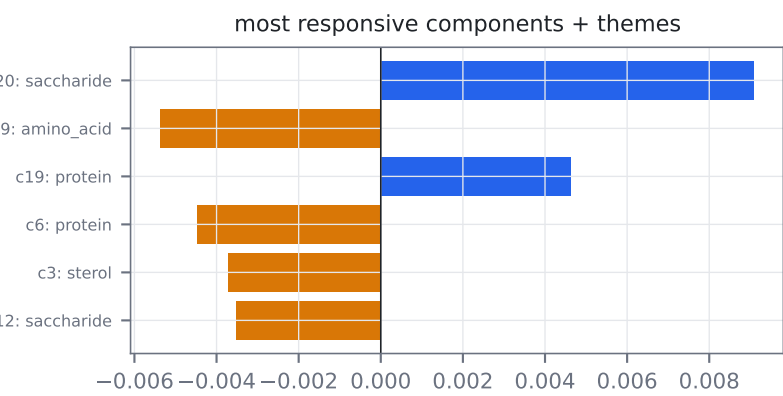
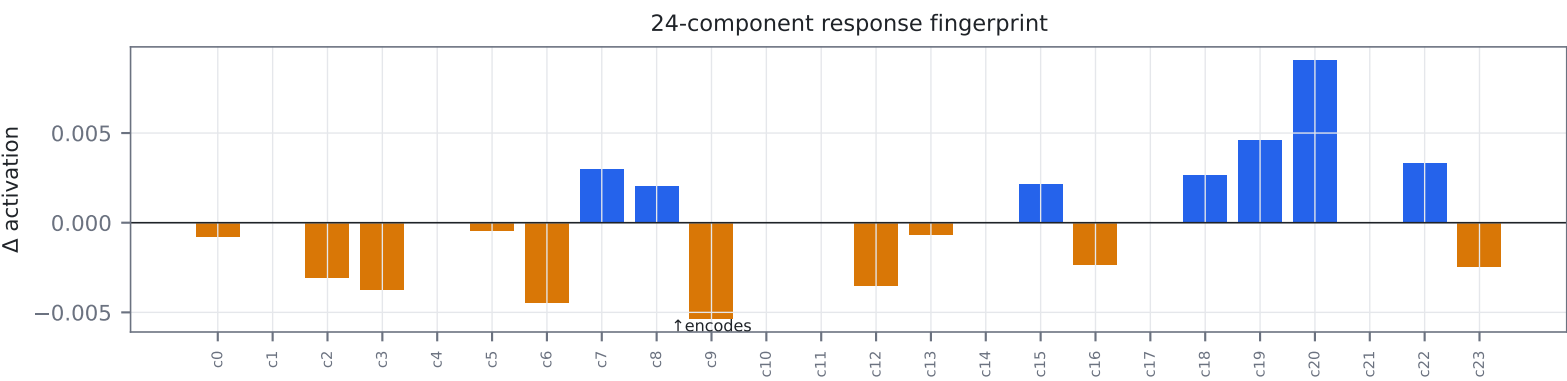
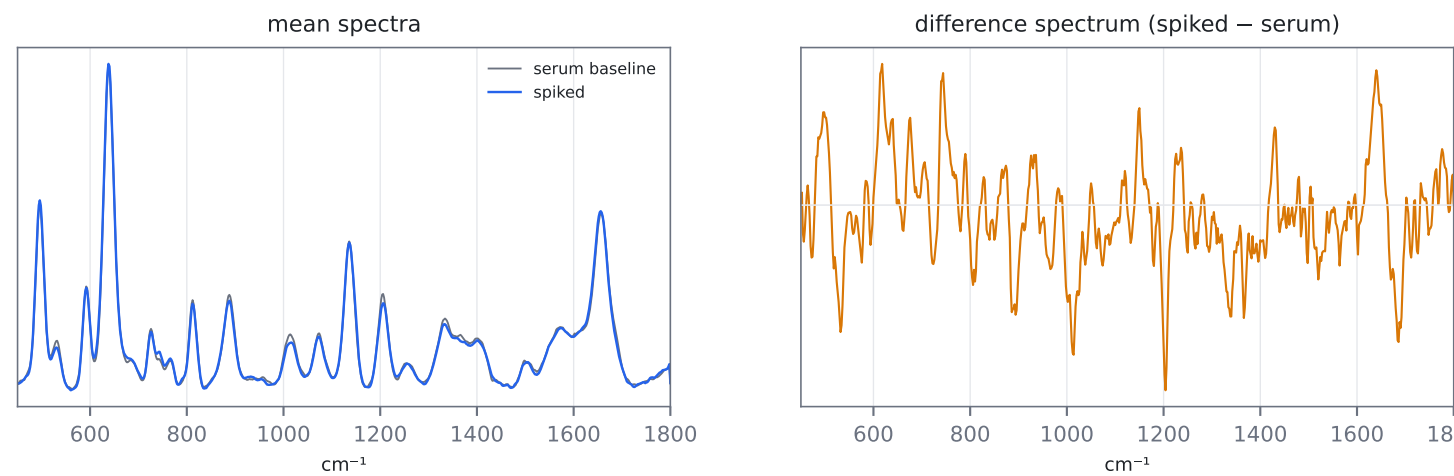


Interpretation. The spike drives components c20, c22, c18 most strongly (themes: saccharide, protein, saccharide). Its own encoding component(s) c11 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is cos -0.21 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

glycine

family: amino_acid · weak / non-responder · spike encodes: c9



nearest neighbours (response cosine)

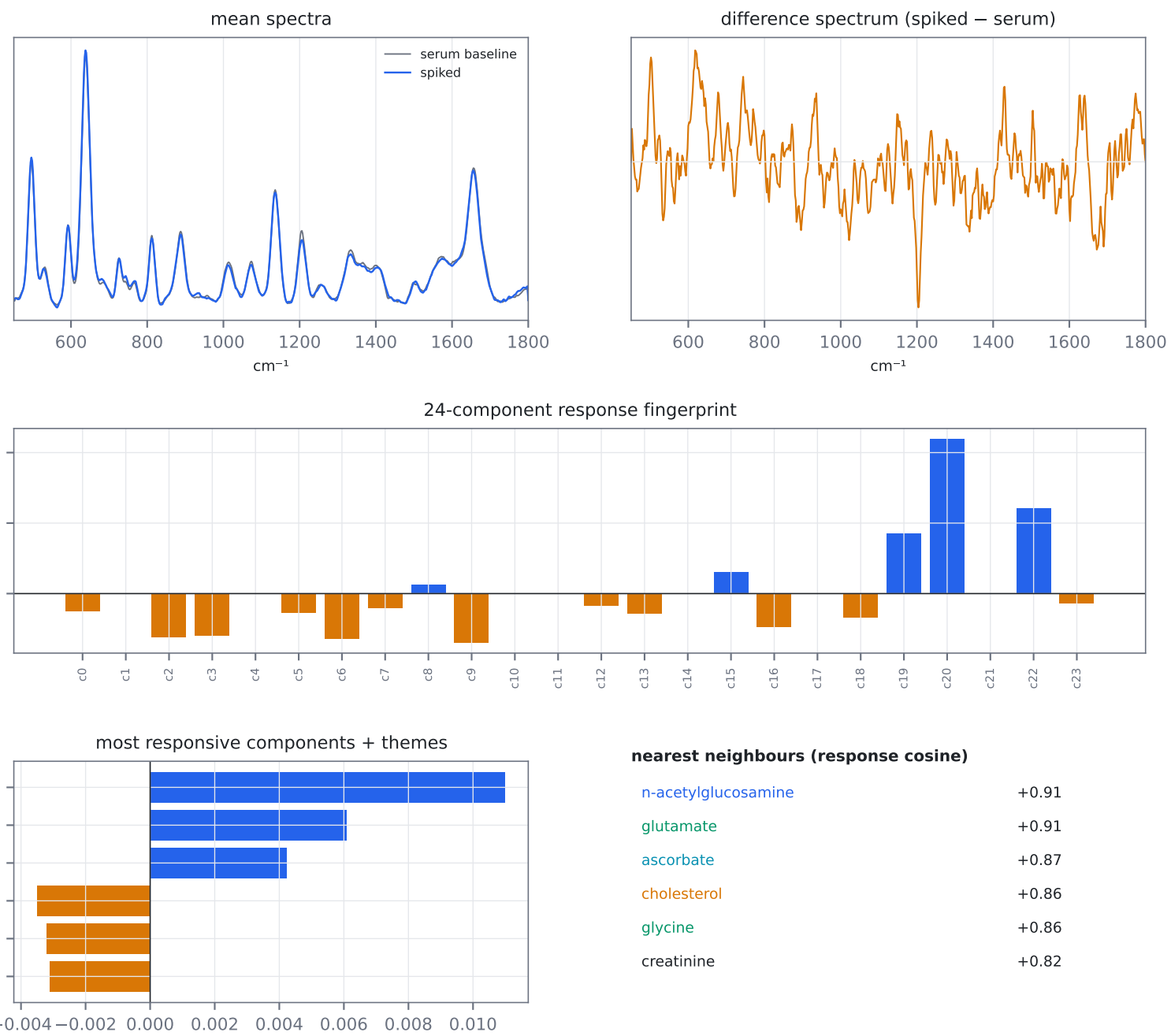
ascorbate	+0.92
glucose	+0.91
leucine	+0.88
cholesterol	+0.88
triolein	+0.86
creatinine	+0.81

Interpretation. The spike drives components c20, c9, c19 most strongly (themes: saccharide, amino_acid, protein). Its own encoding component(s) c9 ARE among the strongest responders — an identity match. Pure-vs-serum consistency for this analyte is cos -0.03 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

triolein

family: lipid · weak / non-responder · spike encodes: no atlas component

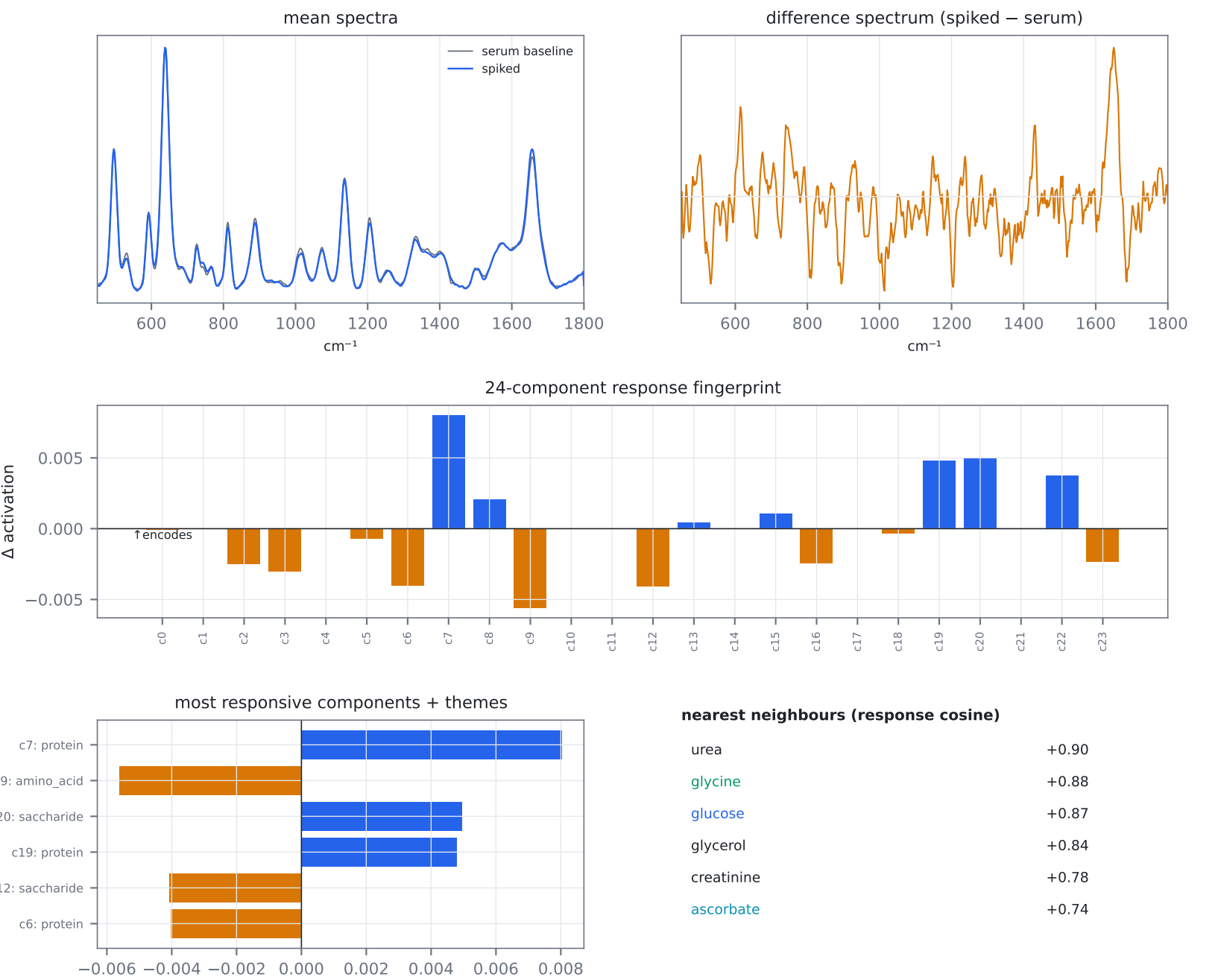


Interpretation. The spike drives components c20, c22, c19 most strongly (themes: saccharide, protein, protein). This analyte does not correspond to any single atlas component, so no identity match is possible. Pure-vs-serum consistency for this analyte is cos +0.10 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

leucine

family: amino_acid · weak / non-responder · spike encodes: c0

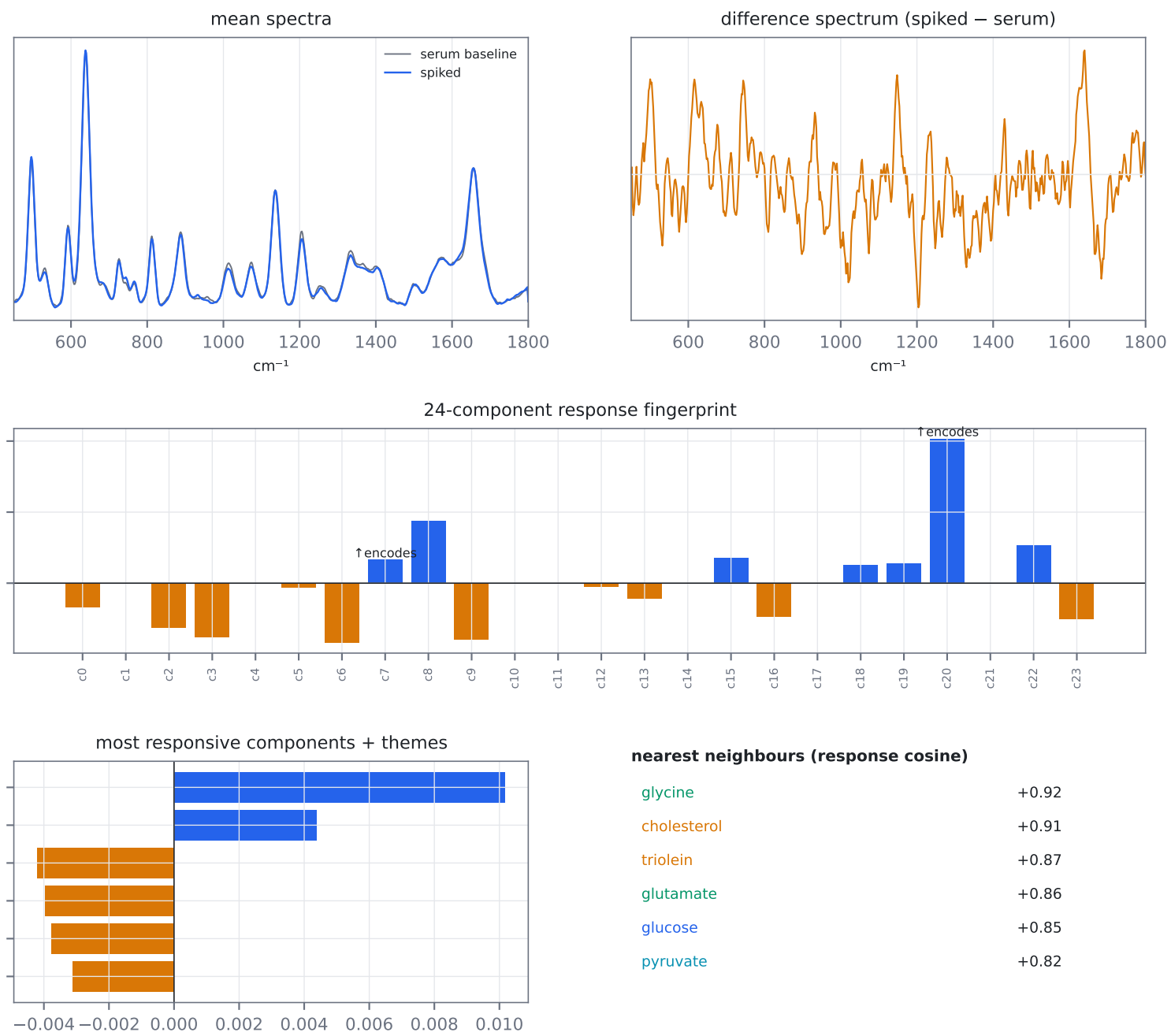


Interpretation. The spike drives components c7, c9, c20 most strongly (themes: protein, amino_acid, saccharide). Its own encoding component(s) c0 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is cos +0.13 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

ascorbate

family: organic_acid · RESPONDER · spike encodes: c7, c20

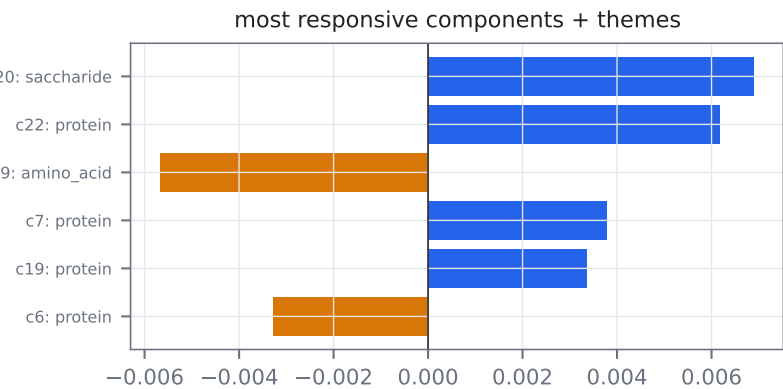
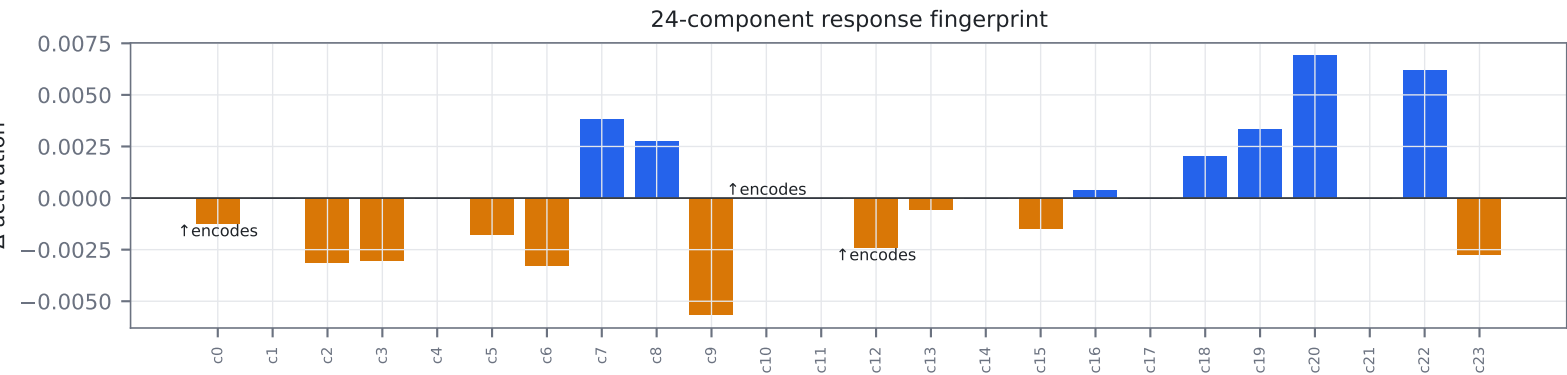
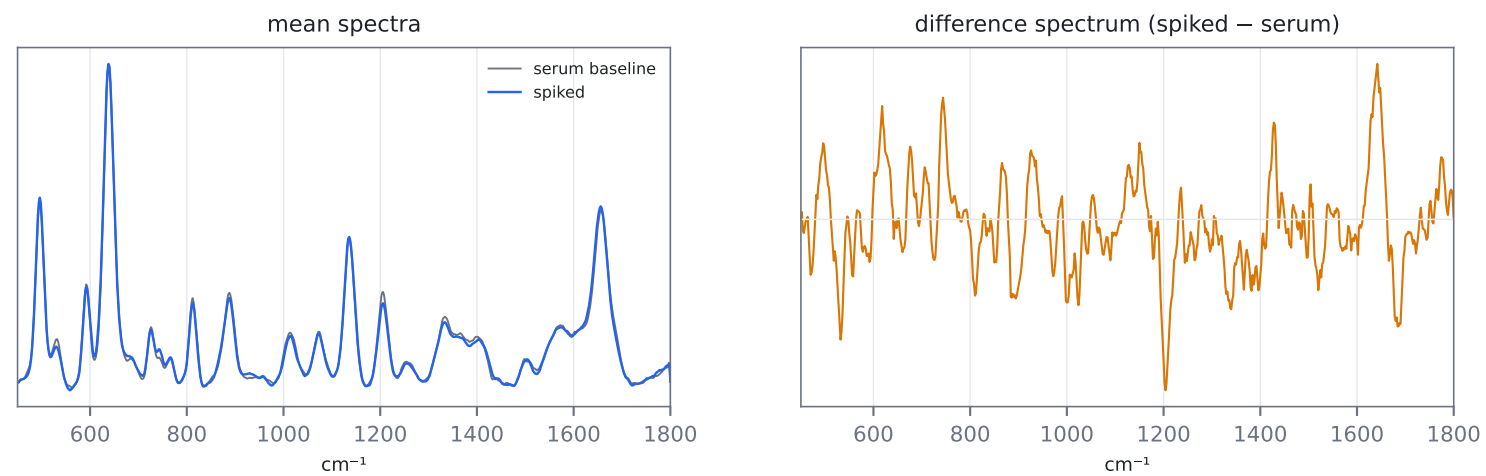


Interpretation. The spike drives components c20, c8, c6 most strongly (themes: saccharide, protein, protein). Its own encoding component(s) c7, c20 ARE among the strongest responders — an identity match. Pure-vs-serum consistency for this analyte is cos +0.48.

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

glucose

family: saccharide · weak / non-responder · spike encodes: c0, c10, c12



nearest neighbours (response cosine)

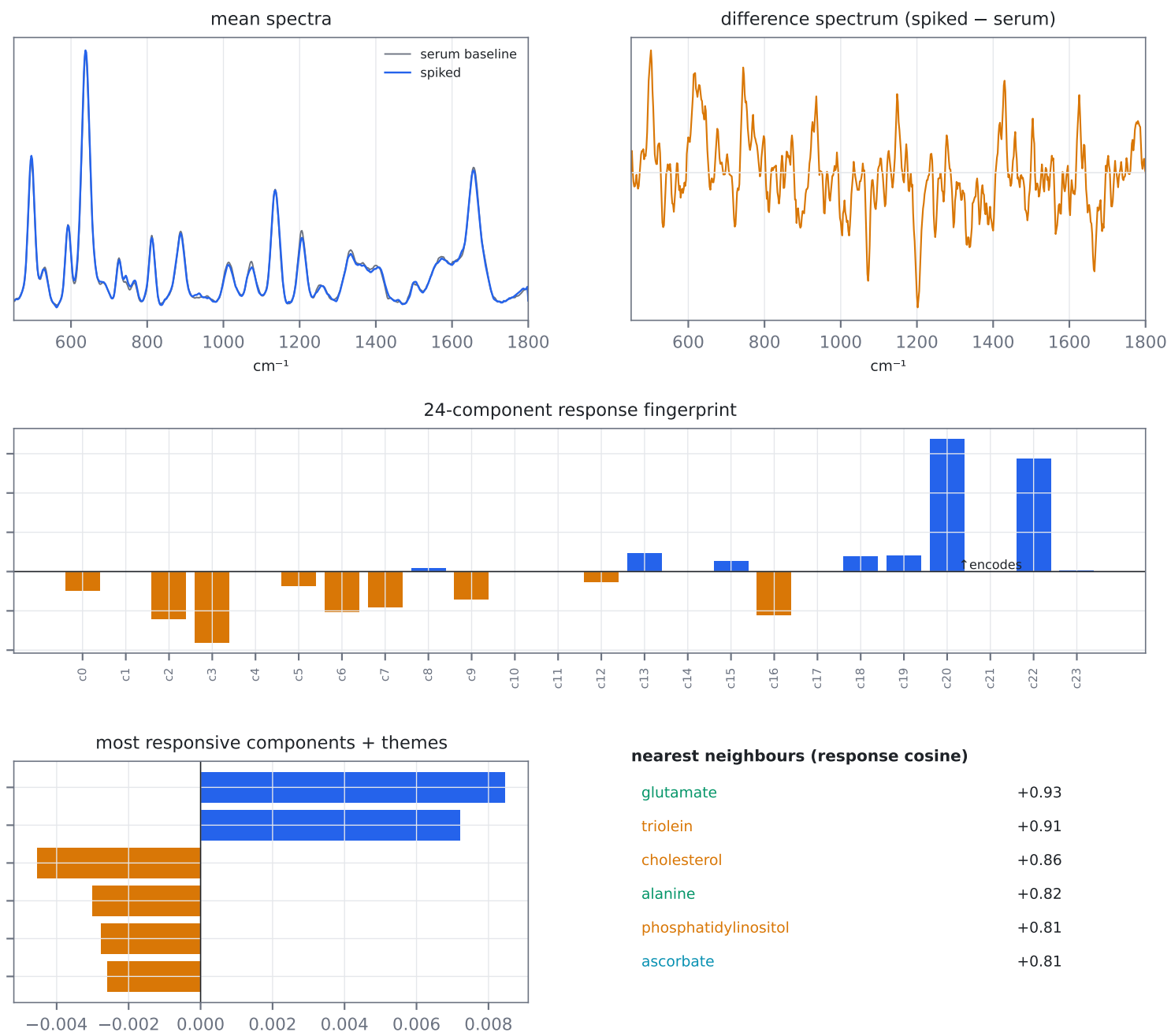
glycine	+0.91
creatinine	+0.90
leucine	+0.87
ascorbate	+0.85
triolein	+0.80
cholesterol	+0.78

Interpretation. The spike drives components c20, c22, c9 most strongly (themes: saccharide, protein, amino_acid). Its own encoding component(s) c0, c10, c12 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is cos -0.26 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

n-acetylglucosamine

family: saccharide · weak / non-responder · spike encodes: c21

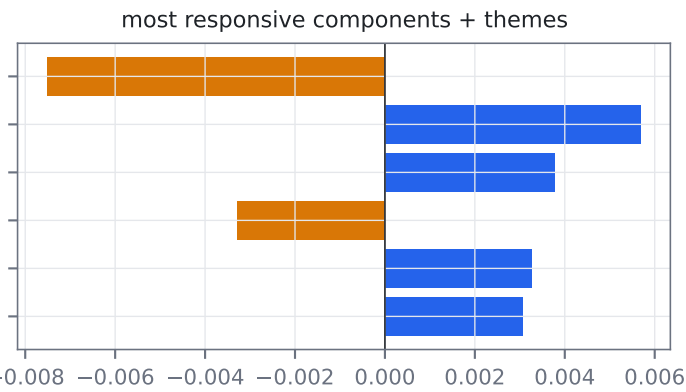
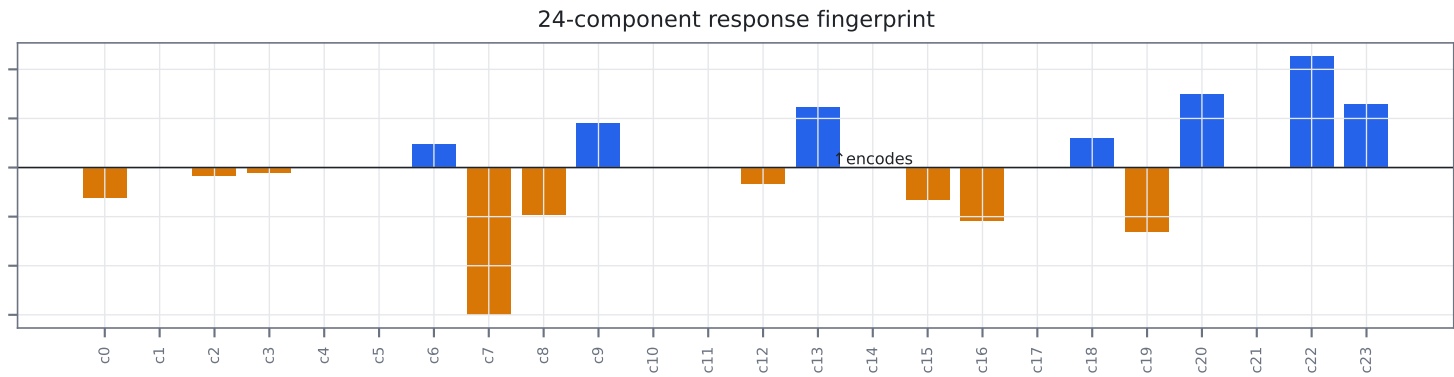
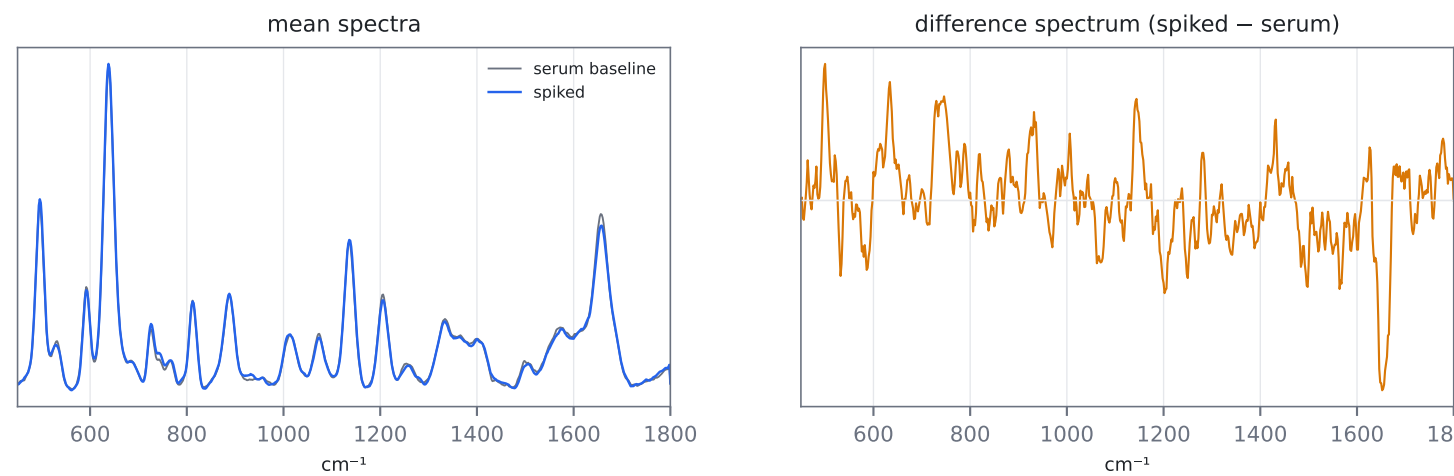


Interpretation. The spike drives components c20, c22, c3 most strongly (themes: saccharide, protein, sterol). Its own encoding component(s) c21 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is $\cos -0.13$ (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

phosphate

family: organic_acid · weak / non-responder · spike encodes: c14



Interpretation. The spike drives components c7, c22, c20 most strongly (themes: protein, protein, saccharide). Its own encoding component(s) c14 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is cos +0.04 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

nearest neighbours (response cosine)

fructose	+0.86
thymine	+0.84
riboflavin	+0.81
fructose-6-phosphate	+0.80
citrate	+0.78
phosphoenolpyruvate	+0.77

urea

family: small_nitrogenous · weak / non-responder · spike encodes: c6



Interpretation. The spike drives components c7, c12, c22 most strongly (themes: protein, saccharide, protein). Its own encoding component(s) c6 are present but not the strongest responders. Pure-vs-serum consistency for this analyte is cos -0.15 (matrix-specific).

A component is a mathematical Raman motif; themes are tentative and never molecular assignments.

Mechanistic interpretation and DART

Part 15 — has the loop closed?

Partially, and at the level of component IDENTITY rather than theme label. For adenine, the single strongest-responding component is c3 — whose top reference analyte IS adenine — in 4 of 6 substrate x laser arms, a reproducible closure across substrates and lasers. For ergothioneine the own-encoding component (c19) responds but is not the strongest, and c19 is a low-purity generic component, so that closure is weak. Overall, the analyte loads its responding component in 7/128 responsive-component instances, concentrated in the strongest responders.

The component AUDIT predicts response identity well but its THEME LABELS predict it poorly. Theme-label agreement is only 15/128, because several low-purity components carry misleading coarse labels (c3 labelled 'sterol' actually encodes adenine; c13 labelled 'pyrimidine' is thymine-dominated). The underlying components are chemically real; the labels are the weak link.

It supports the ontology's HIGH-CONFIDENCE, chemically-focused entries and challenges the coarse labels on low-purity entries. The purine sub-structure recovered from perturbation (two anti-correlated pairs) is finer than the single 'purine' ontology axis.

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components that GAIN identity confidence: [3]
components whose LABEL loses confidence: [3]
```

Components needing reinterpretation

c3: labelled 'sterol', driven by adenine, top reference ['adenine', 'acetyl coenzyme a', 'estrone', 'coenzyme a']
Reading: perturbation fingerprints split the purines into {adenine, hypoxanthine} (6-amino/oxo) and {xanthine, guanine} (2,6-dioxo/oxo-amino), anti-correlated across the divide — a chemically correct sub-classification the atlas was never told about

Reading: enzymatic urate depletion produces a SELECTIVE loss in a purine-encoding component rather than a global change — mechanistically consistent with removing a purine

Pure-vs-spike consistency is near zero (median cos -0.012): an analyte does NOT in general activate the same components in serum as in buffer, so these fingerprints are matrix-specific and must not be treated as matrix-invariant signatures.

Part 16 — implications for DART (grounded)

A DART experiment applies a controlled electrochemical perturbation and records a time/potential series of spectra. Projected into the frozen atlas, each such series is a TRAJECTORY of the same type this study built for chemical dose series.

- A trajectory-fingerprint schema (path length, straightness, curvature, component turnover, response entropy, OOD evolution) already computed for 7 chemical dose trajectories — the first reference library a DART trajectory could be compared against.
- Evidence that dose trajectories are monotonic and saturating and that the strongest responding component often matches analyte identity — so a DART trajectory that redox-cycles a known species has a concrete, testable expectation (motion along that species' component).

Recommended first DART test: Redox-cycle a single strong Ag adsorber with a known atlas component (e.g. a purine) and check whether the projected trajectory moves predominantly along that component and returns toward baseline on the reverse sweep — a direct falsifiable prediction.

limitation: Every dataset here is Ag/Au-SERS projected into a Raman atlas and is out of domain; a DART measurement in yet another modal
limitation: Pure-vs-serum consistency is ~0, so a component response is matrix-specific; a DART trajectory in an electrochemical cell
limitation: Theme labels on low-purity components are unreliable; DART interpretation must anchor on component IDENTITY (reference load)
This study does not demonstrate DART compatibility; it provides the trajectory vocabulary and one falsifiable prediction for a future DART experiment.